My Little Music Grammar

LEVEL 2



I read, listen to and compose music

MUSIC MOTION

My Little Music Grammar · Level 2
I read, listen to and compose music
Pupil's workbook for the *Music in the Classroom* programme

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CANELO SAYS

Every chapter begins on a right-hand page. And at the end of the journey a diploma is waiting for you: I've composed my opera!

For teachers and families

What is in this second workbook

In Level 1 the children learnt to read: the stave, the note values, the bar. In Level 2 reading becomes a journey. Each chapter visits a period of music history, with a work to listen to while following the score: Mozart, Schubert, Bach, Bartók, Ravel, Beethoven… and the Song of Seikilos, which they already know.

At the start, 'The journey of sounds' shows the whole route, from the clay tablets of Mesopotamia to today, and the child ticks off each stop. At the end of the year the class composes and performs a little opera: **'I've Lost My Sandwich!'** Everything learnt in the workbook goes into writing it.



Each chapter has a **Listen along** box, with the work from that period, and an **At home** box: ten minutes a week, with the computer keyboard on **musicmotionsuite.com**. You don't need to know any music to support your child.

How each chapter is built

- **The story**: the new idea, told with a picture.
- **Let's play!**: little pencil tasks to do in the workbook.
- **Listen along**: the work from the time machine.
- **At home**: a ten-minute game with the family.

Each chapter goes with a lesson in the teacher's guide of **Music in the Classroom**, on musicmotionsuite.com: what the child does in the workbook, the class has already played in the lesson.

BEFORE WE BEGIN

The journey of sounds

Intervals existed before writing

Long before anyone wrote down a note, people were already singing and playing with the same intervals you are going to learn: the octave, the fifth, the fourth. Nobody invented them: strings, pipes and the voice found them. This workbook follows their journey around the world and through time. When you reach a stop, put a ✓ next to it.



☐	**around 1800 BC** **Mesopotamia**	Clay tablets with instructions for tuning the lyre: the first ‘scores’ are recipes for tuning. → ch. 13
☐	**about 3000 years ago** **India**	The hymns of the Sama Veda are kept by heart, recited forwards and backwards. The notes are called Sa, Re, Ga, Ma, Pa, Dha, Ni. → ch. 6
☐	**around 1200 BC** **China**	Twelve bamboo pipes, tuned one after another in fifths: the Lülü system. → ch. 4
☐	**centuries ago** **The Americas**	The quena, the pan pipes (zampoña) and the pututu keep the octave in the length of their pipes. → Level 1

☐	**6th century BC – 1st century AD** **Greece**	Pythagoras measures strings with numbers; Seikilos carves his song on a column. → ch. 5, 6, 7 and 13
☐	**9th to 11th centuries** **The Arab world**	Al-Kindi, Al-Farabi and Ibn Sina translate, preserve and enrich Greek knowledge.
☐	**9th to 11th centuries** **The monasteries of Europe**	The monks write neumes, then lines: music writing is born. → ch. 7 and 9
☐	**Middle Ages** **The first keyboard**	Twelve keys in every octave: the map of sounds under your hand. → ch. 4 and 5
☐	**1720 and 1723** **Bach**	The little music book for his son and the 15 Sinfonias, each in its own key. → ch. 3 and 8
☐	**1783 and 1792** **Mozart**	The 'Alla Turca' and the dice game. → ch. 1 and 12
☐	**1824** **Beethoven**	The Ninth and the 'Ode an die Freude' *(Ode to Joy)*. → ch. 9
☐	**1825** **Schubert**	The 'Ave Maria'. → ch. 2
☐	**1906** **Bartók**	Records the songs of the villages with a phonograph. → ch. 5
☐	**1931** **Ravel**	The Concerto en sol *(Piano Concerto in G)*. → ch. 6
☐	**today** **Your class**	Composes its own opera and performs it for the first time: 'I've Lost My Sandwich!' → ch. 12



CANELO SAYS

Strings, pipes and the voice were always the teachers. And I'll come with you on the whole journey! You don't need to learn the dates by heart: they are like the signposts along the road, so you know where we are.

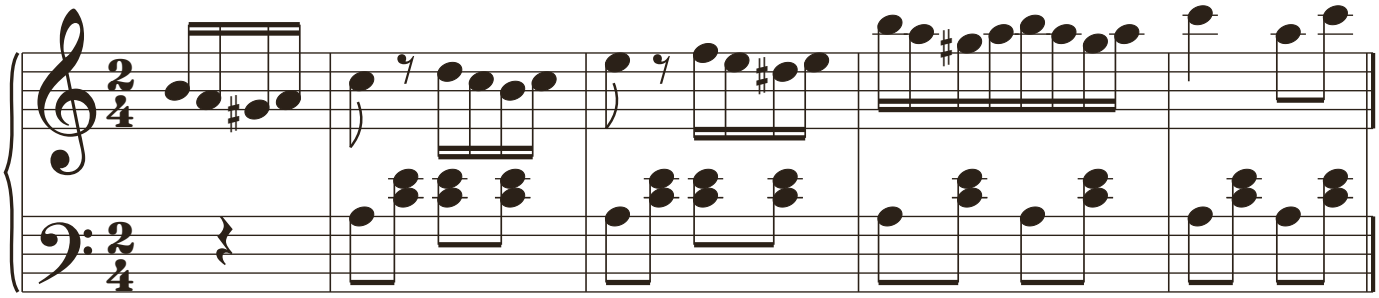
CHAPTER 1

The route through a piece of music

Level 1 revision · the upbeat, the repeats and the Coda



Wolfgang Amadeus Mozart · Piano Sonata in A major, KV 331 · III. Rondo 'Alla Turca' (Vienna, 1783)

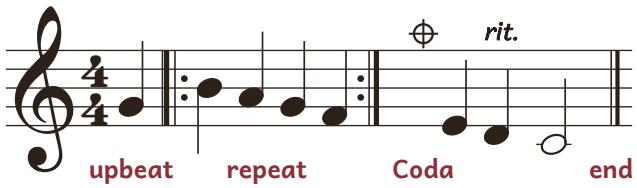


*The first bars. The piece begins with an upbeat: four quick notes that together last one beat. Listen to it on **musicmotionsuite.com**, in the Keyboard demos: 'Alla Turca'.*

You can already read: the notes live on the lines and in the spaces, the treble clef curls around line 2, the note values are pieces of a bar of chocolate and the time signature tells you how many beats fit between two bar lines.

The **upbeat** is not a complete bar: here it is four quick notes, one beat altogether. Usually the last bar completes it: it is missing exactly what the upbeat has, so it all adds up when you repeat the verse.

Now we are going to look at the **route** through a piece of music. You don't always read a score straight through, from the first note to the last: it has signs on it, like a map. Sometimes the piece begins **before** the first complete bar (that is the **upbeat**, or anacrusis). Some parts are played twice (the **repeats**) and there is a special ending, the **Coda**: that is Italian for 'tail'.



The signposts: upbeat, repeat signs, Coda and ritardando (slower and slower).



CANELO SAYS

The two little dots next to the bar line are like an arrow: 'Go back and play it again!'

Listen along · Vienna, 1783

Mozart writes a piano sonata. The last movement, **'Alla Turca'**, imitates the Turkish military bands of the Janissaries, with cymbals and drums, which were fashionable in Vienna at the time. A refrain keeps coming back: that is called a **rondo**.

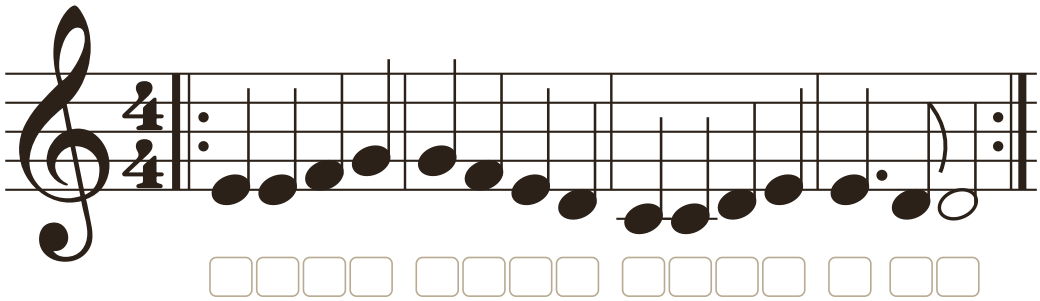
The words you already know

From the little dictionary in Level 1, 'Music words'. With them you can already read the 'Ode an die Freude'.

- **Beat**: the heartbeat of the music. Always evenly spaced, like the tick-tock of a clock.
- **Stave**: the house of the notes: five lines and four spaces.
- **Line and space**: where each note lives: with a line through it, or between two lines.
- **Treble clef**: the key to the house: it gives line 2 the name G.
- **Note value**: the shape of the note. It tells you how long it lasts.
- **Semibreve, minim and crotchet**: 4, 2 and 1 beat: the whole bar of chocolate, half of it and a quarter.
- **Quaver**: half a beat. Two quavers are said 'ti-ti'.
- **Rest**: the time when the music is silent. The beat keeps counting.
- **Bar**: a group of beats between two bar lines. The first is the strong beat.
- **Time signature**: the two numbers: on top, how many beats; underneath, which note value is the beat.
- **Tempo**: how fast the beat goes. At 60: one beat per second.

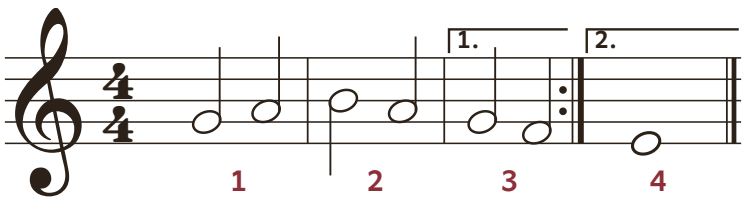
Let's play!

1. This is Beethoven's 'Ode an die Freude'. Write the name of each note underneath it.



2. Circle the two repeat signs: the one at the beginning and the one at the end.
3. How many bars does the piece have? _____ And if you repeat everything? _____
4. A section of 8 bars is between repeat signs. How many bars do you hear? _____
5. 'Alla Turca' is in 2/4 and begins with an upbeat of 1 beat. The last bar completes the upbeat. How many beats does that bar have? _____
6. **Challenge:** the first two sections of 'Alla Turca' are 8 and 16 bars long, and each section is played twice. How many bars do you hear altogether? _____
7. **The dictation:** your teacher plays a rhythm of four beats (crotchets, quavers, a minim, a rest). Listen, say it with syllables and write it here. There are two rounds!

8. **1st- and 2nd-time bars:** the first time, you play the 1st-time bar; on the repeat, you skip the 1st-time bar and play the 2nd-time bar. Write down, in order, the numbers of the bars you hear.
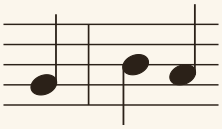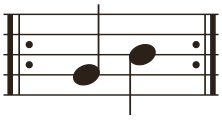
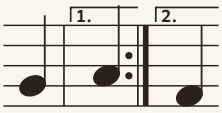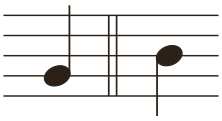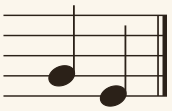
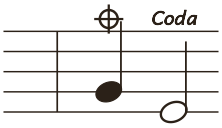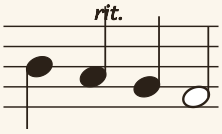


_____ _____ _____ _____ _____ _____

The signposts in the score

To look things up all year

You don't always read a score simply from left to right: it has signs that tell you which way to go, like the signposts along a road.

What it looks like	What it's called	What it says
	**Upbeat**	The notes that come before the first complete bar. In 'Alla Turca' they are four quick notes that together last one beat.
	**Repeat signs**	Whatever is between the two signs, you play twice. If there is no sign at the beginning, you go back to the start of the piece.
	**1st- and 2nd-time bars**	The first time, you play the 1st-time bar; on the repeat, you jump to the 2nd-time bar.
	**Double bar line**	One section ends and another begins.
	**Final bar line**	A thin line and a thick line: the piece is over.
	**Coda**	Italian for 'tail': the special ending of the piece, which is heard only once.
	**rit.** (ritardando)	Slower and slower, little by little.

Later on you will also meet **D.C.** (*da capo*, 'from the head': back to the beginning) and **Fine** ('end': where you stop after going back).

At home

Listen together to 'Mozart Alla Turca'. The child puts up a hand every time the march with the cymbals comes back.

On the keyboard: play the 'Ode an die Freude' at musicmotionsuite.com/apps/mmk-demo.html, with the keys C C V B · B V C X · Z Z X C · C X X.

CHAPTER 2

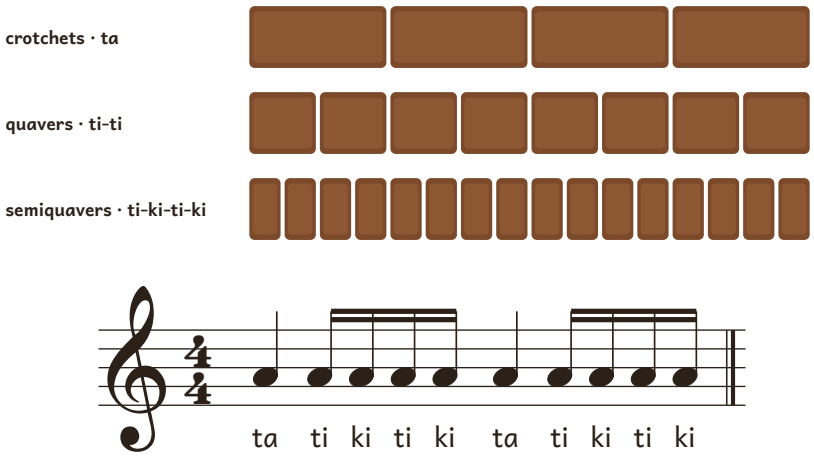
One and a half pieces

Semiquavers and the dot

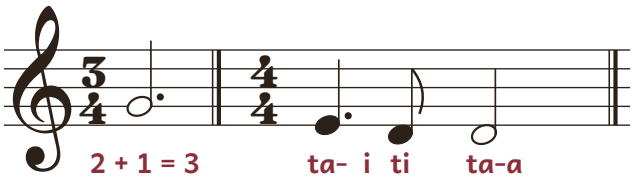


You break the bar of chocolate into 4, into 8 and into 16 pieces.

In Level 1 the bar of chocolate was broken into as many as eight pieces: the quavers. If you break each quaver in half again, you get **semiquavers**: four on every beat. You write them with **two beams** and you say 'ti-ki-ti-ki'.



A **dot** after a note makes it **half as long again**. The minim lasts 2 beats; with a dot, 2 + 1 = 3. The crotchet lasts 1 beat; with a dot, 1 + ½. Almost always a quaver follows to fill up the two beats: long-short, like a galloping horse. 'Ta-i ti'.



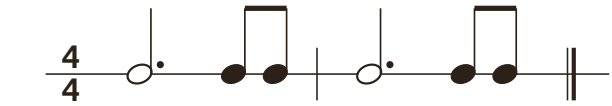
The end of the Ode: the dotted crotchet with its quaver. 'Ta-i ti, ta-a'.

Let's play!

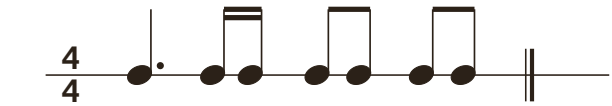
1. Match with an arrow:
   dotted minim → _____ beats · dotted crotchet → _____ beats
2. Find words with four syllables (wa-ter-mel-on, cat-er-pil-lar) and write them here. Each word fits into one beat: that is four semiquavers.

---

3. Fill the 3/4 bar with one single note value: _____
4. **This is how the voice sings in the 'Ave Maria'.** Read each rhythm with syllables and circle the dotted notes. How many beats does each one last?



bars 5 and 6



bar 10

Listen along · Vienna, 1825

Franz Schubert sets a poem to music: a girl, hidden in a cave, is praying. It is the **'Ave Maria'**, one of the most sung melodies in the world. In the voice there are dotted crotchets that let it breathe; in the piano, quick notes that sound like a harp. It is **Romantic** music: what matters is what you feel.

At home

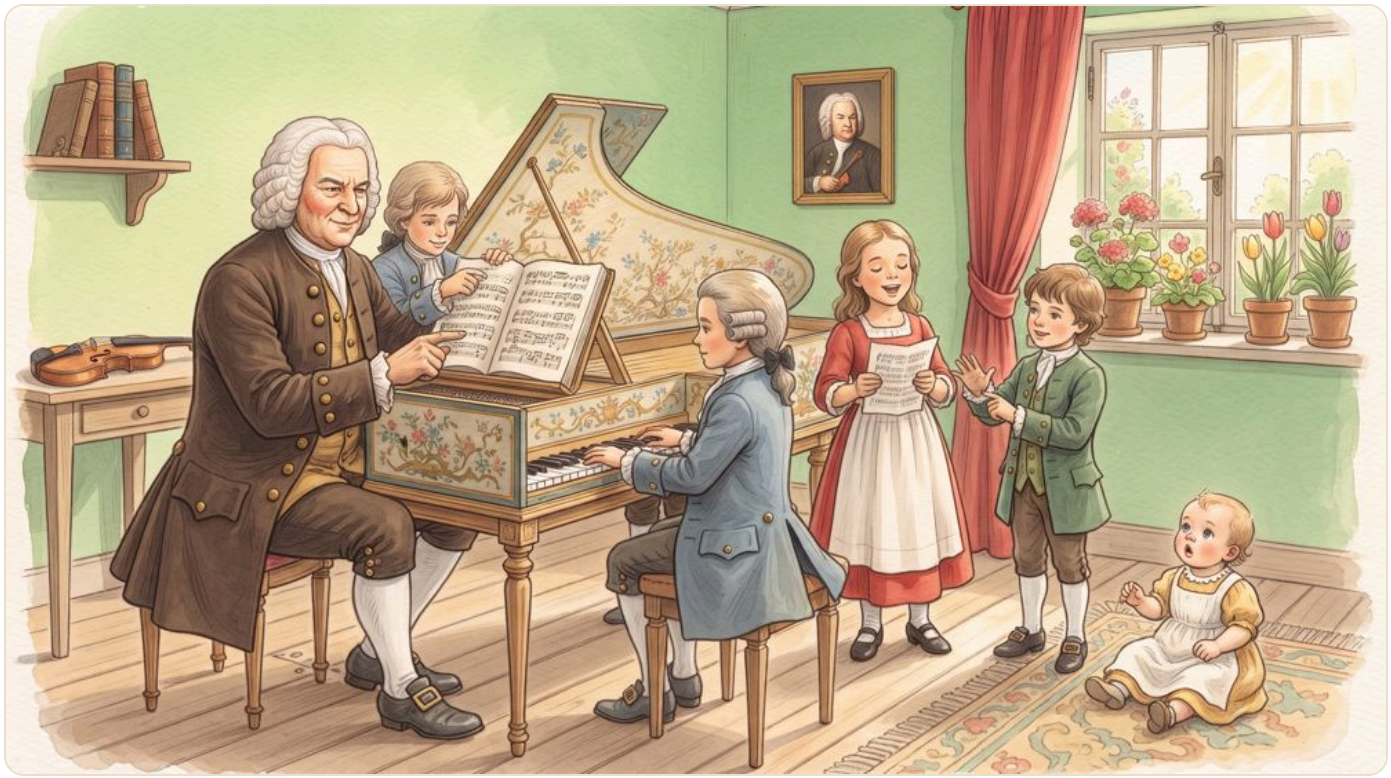
The gallop: one person says 'ta-i ti, ta-i ti' and the other gallops around the house, a long step and a short step. Then swap. Listen together to the 'Ave Maria': is it music for marching or for calming down?

Something wilder: listen to the last movement (Presto) of Schubert's *String Quartet in D minor*: the strings gallop like horses!

CHAPTER 3
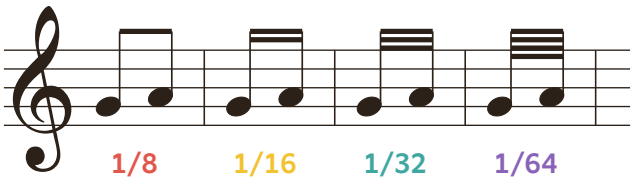
The fast notes

The demisemiquaver, the hemidemisemiquaver, the triplet and ornaments



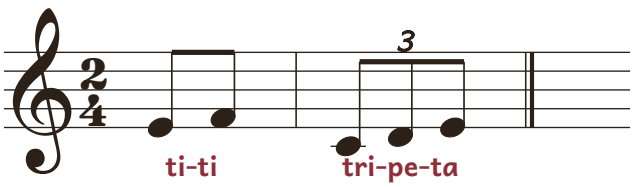
Bach gives his children music lessons at home.

We keep breaking up the chocolate bar. Half a semiquaver is a **demisemiquaver** (a 32nd of a semibreve), and half of that is a **hemidemisemiquaver** (a 64th of a semibreve). Each time a note lasts half as long, it gets **one more beam**. A crotchet holds 2 quavers, 4 semiquavers, 8 demisemiquavers or 16 hemidemisemiquavers.



One, two, three and four beams: quavers, semiquavers, demisemiquavers and hemidemisemiquavers.

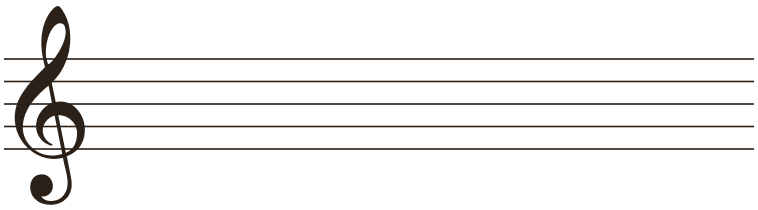
Sometimes you want **three** notes in the time of two. Then you write three quavers with a 3 above them: that's a **triplet**. You say 'tri-pe-ta' on one beat.



Two quavers, 'ti-ti', and a triplet, 'tri-pe-ta': both last one beat.

Let's play!

1. How many beams does it have? Quaver _____ · semiquaver _____ · demisemiquaver _____ · hemidemisemiquaver _____
2. Fill in the gap: a crotchet holds _____ demisemiquavers.
3. Draw a triplet on the stave, with the notes C, D, E.





CANELO SAYS

Do you know what a trill sounds like? Like when I shake the water off me: quick, quick, side to side!

Did you know...? Ornaments, such as the trill and the mordent, are fast notes that decorate a tune. In Bach's time, people who could ornament well were greatly admired.

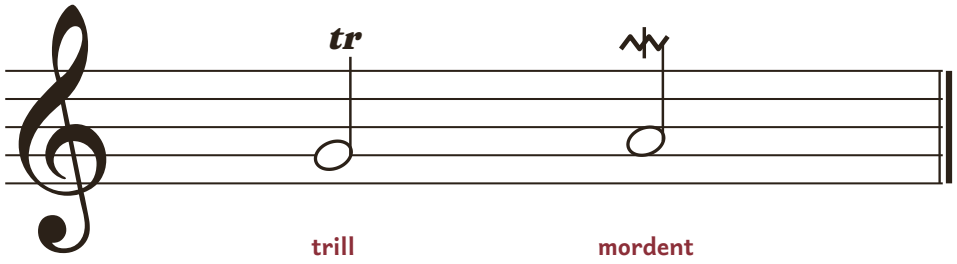
Let's listen · Germany, 1720

Johann Sebastian Bach writes a notebook for his nine-year-old son Wilhelm Friedemann, to learn to play the keyboard, with a table of ornaments. Years later he writes the **15 Sinfonias** for his pupils. Listen to **No. 9, in F minor**, and look out for the ornaments.

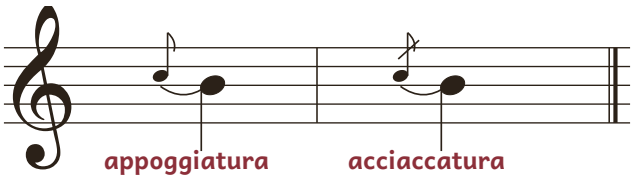
At home

- **The chocolate bar:** together, break a bar of chocolate (or a sheet of paper) into halves, quarters, eighths... How far can you go? The child says the name of each note value.
- **Bach twice:** listen to Sinfonia No. 9 on musicmotionsuite.com, in the Score, and then in the Keyboard, in Tempered. Can you hear the difference?
- **The trill:** the child plays a trill on the computer keyboard, switching very quickly between two neighbouring keys (G and H).
- **Tri-pe-ta:** say 'tri-pe-ta' and 'ti-ti' together, one beat each. Which one fits in the same clap? Both!

Ornaments are written with small signs above the note. The **trill**, 'tr', switches very quickly between the written note and the one above it. The **mordent**, a short zigzag, makes a lightning-fast trip to the neighbouring note and comes back.

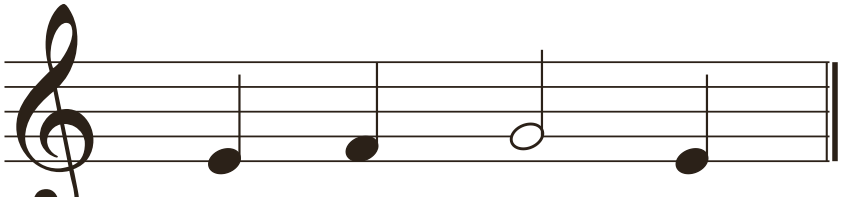


The **appoggiatura** is a small note before the main note, which leans on it. If it has a little slash through it, you play it as fast as possible. 'Alla Turca' is full of them!



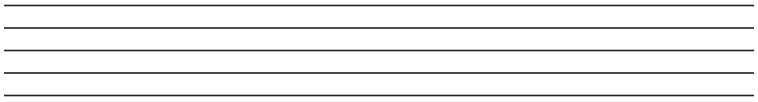
Let's play!

1. Write 'tr' above the longest note.



2. Which ornament switches very quickly between two notes? _____

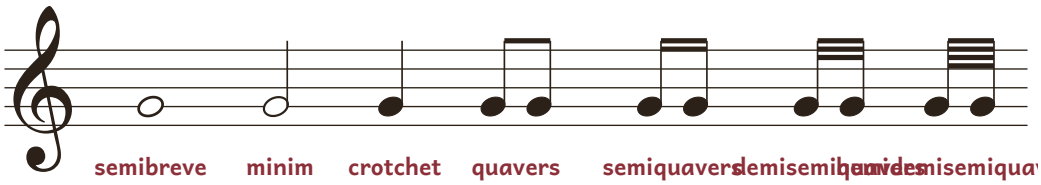
3. **Your 2/4 bar:** write a triplet or demisemiquavers, and a crotchet with a trill.
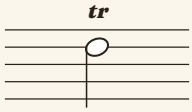


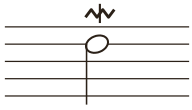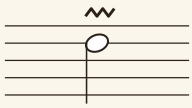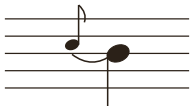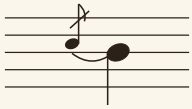
4. **Challenge:** how many hemidemisemiquavers fit into a semibreve? _____

Did you know...? In class we listen to Bach's Sinfonia No. 9 in three tunings: **tonal** tuning, **Werckmeister** tuning, which was used in his day, and the **tempered** tuning of today's piano.

Note values and ornaments



Note value	Part of a semibreve	Beats	Beams
Semibreve	1 whole	4	—
Minim	1/2	2	—
Crotchet	1/4	1	—
Quaver	1/8	½	1
Semiquaver	1/16	¼	2
Demisemiquaver	1/32	1/8	3
Hemidemisemiquaver	1/64	1/16	4

What it looks like	What it's called	What it does
	**Trill**	The note and its upper neighbour, one after the other, very fast.
	**Mordent**	The note, its lower neighbour and the note again, very fast.
	**Upper mordent**	The same, but with the upper neighbour.
	**Appoggiatura**	A small note before the main note, which leans on it.
	**Acciaccatura**	An appoggiatura with a little slash through it: you play it as fast as possible.



CANELO SAYS

When I scratch behind my ear with my paw, I go as fast as hemidemisemiquavers: scritch-scritch-scritch-scritch!

REVISION · CHAPTERS 1 TO 3

Don Quixote sets off

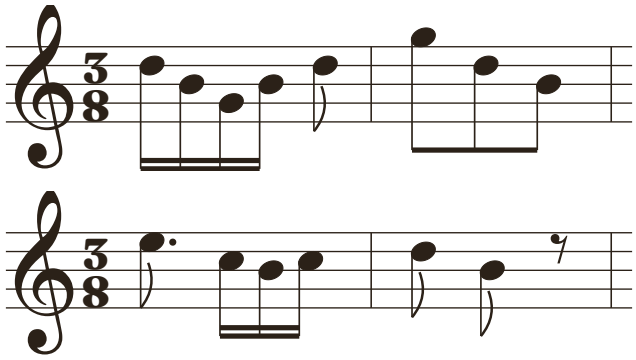
A book, some music and the note values you already know



Don Quixote on Rocinante and Sancho Panza on his donkey, crossing the fields of La Mancha.

In 1605, in Spain, Miguel de Cervantes published the story of **Don Quixote**, a gentleman who read so many books about knights that he set off in search of adventure, in an old suit of armour, riding **Rocinante**, a terribly skinny horse. His neighbour **Sancho Panza** goes with him, on his donkey.

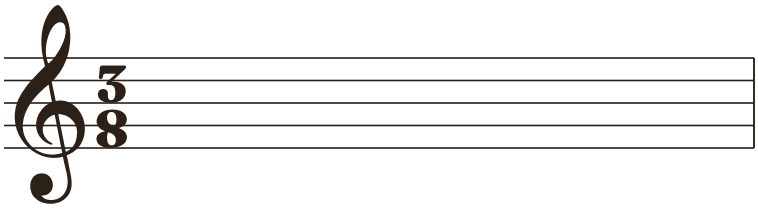
More than a hundred years later, in Germany, **Georg Philipp Telemann** turned the book into music: a suite for strings in which each piece tells an adventure. Every bar has **three quavers**. This is how they gallop:



Above, Rocinante's gallop; below, Sancho's donkey.

Let's play!

1. How many semiquavers are there in Rocinante's first bar? _____ How many quavers are they worth together? _____
2. The dotted quaver of Sancho's donkey is worth _____ semiquavers.
3. Write your own galloping bar, with note values that add up to three quavers.



4. **Challenge:** Telemann wrote music for Don Quixote's attack on the windmills. What other adventure would you like to hear? _____

Let's dance!

Make up a dance together. To Rocinante's gallop, the knights jump lightly: 'ta-ka-ta-ka-tan'. To Sancho's donkey, the squires walk at their own pace: one long step and three short ones. When the music changes, the steps change.



CANELO SAYS

I can gallop too! When I run after my ball, my paws go 'ta-ka-ta-ka-tan', just like Rocinante.

Did you know...? Literature and music have gone hand in hand for centuries. After Telemann, Don Quixote went on inspiring operas, ballets and songs, right up to today.

Let's listen · Germany, 18th century

Listen to Telemann's suite *Burlesque de Quixotte*. Find 'Le Galope de Rosinante' and, straight after it, Sancho's donkey. Listen to the attack on the windmills too: can you hear the sails turning?

You can download the score (.mmf) in a .zip from **musicmotionsuite.com/mmf.html**, to follow it in the Score.

CHAPTER 4

Tone and semitone: piano and voice

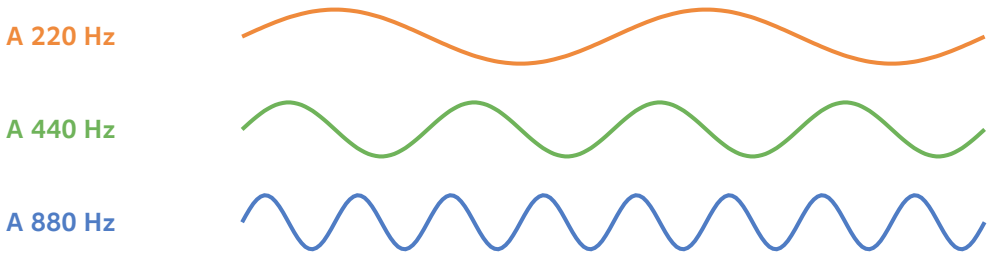
How many times a second · three staircases that live side by side



Sound is something that vibrates: you can feel it, see it and measure it.

Every sound is something that **vibrates**: a string, the air inside a flute, your voice. The **hertz** (Hz) counts how many times that vibration goes back and forth in one second. The A of the tuner is **440 Hz**. The lowest string of the double bass vibrates about 41 times a second: you can see it shaking!

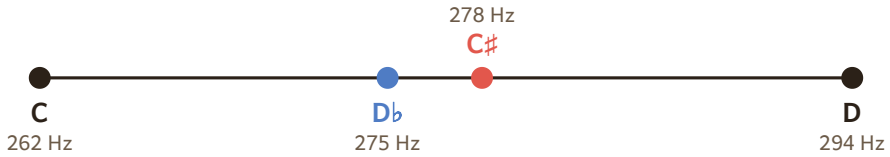
More vibrations, a higher note. And twice as many vibrations is the same note an octave higher: A 220, A 440, A 880.



The same note, A, in three octaves: each time twice as many vibrations.

In 'The Journey of Sounds', Mr Mario tells the children what our ancestors discovered: in Mesopotamia they tuned the lyre with recipes, and in Greece Pythagoras measured the string. Half the string sounds an octave higher: it vibrates twice as fast!

Musicians use **three staircases**. The **tempered** one, the piano's: twelve equal steps, and C♯ and D♭ share the same key. The **tonal** one, the staircase of the voice and the violin: there C♯ and D♭ are two different notes, and C♯ is a tiny bit higher. And the staircase of the **harmonics**, the natural trumpet's. None of them is better: all three are heard every day in the orchestra.



In the voice and on the violin, C♯ is a touch higher than D♭.

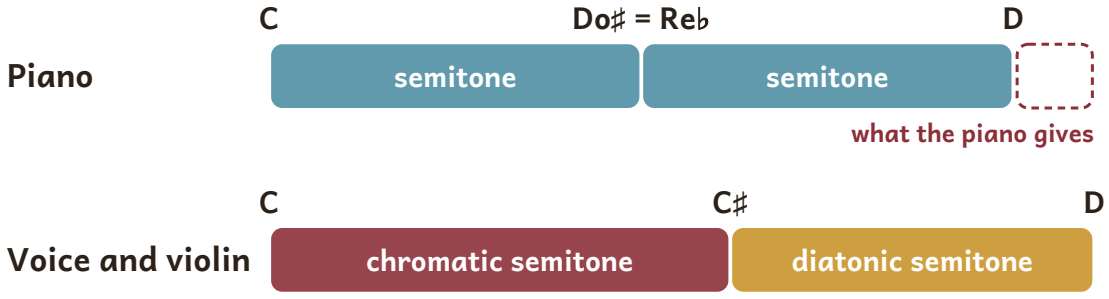
Every step has its name

On the piano, the guitar and electronic keyboards, the smallest step is the **semitone**: they are all the same size.

In the voice and on the violin, the small step is also a **semitone**, but not the same as on the piano: it always has a surname:

- **chromatic**, when the note keeps its name: C → C♯;
- **diatonic**, when the note changes its name: C♯ → D.

A semitone on the piano and a semitone in the voice are not the same thing. — Mr Mario



The real difference is much smaller: here it is enlarged so you can see it.

From C to D: on the piano, two equal semitones and a tone that is a tiny bit shorter; in the voice and on the violin, a chromatic and a diatonic semitone, each a different size.

Why is the piano's tone shorter? To fit **twelve equal semitones** into one octave, the tempered tone gives up a tiny bit of space. The tone of the voice and the violin, which comes from the laws of physics, is a tiny bit bigger. If you add up those little pieces across an octave, together they make a very small difference: the **comma**. You'll get to know it better in Chapter 5.

Let's play!

1. If an A vibrates 440 times a second, the A an octave higher vibrates _____ times.
2. Which is higher in the voice and on the violin: C♯ or D♭? _____
3. Put your hand on your throat and sing a low note and a high note. Which one vibrates faster? _____
4. On the piano, from E to F there is a _____. In the voice, from E to F there is a _____ semitone.
5. Chromatic or diatonic? F → F♯ _____ · F♯ → G _____ · B → C _____

Let's listen · Which is which?

In the Keyboard, the **recs** panel has Bartók on two staircases: **Tonal** and **Tempered**. The difference is small and hard to hear, even for musicians! Without looking, listen to a little bit of each and guess which one is tonal. Then look at the keyboard: if the **cyan strips** light up, it's the tempered one; if the whole keys and the sharp zones light up, it's the tonal one. Your eyes help your ears.

At home

Open Music Motion's free tuner, the **Pythagorean Tuner** (**tuner.mariomaurano.com**), on the phone or the computer. On the computer keyboard, play C♯ (key 1) and D♭ (Shift + X). The tuner will show about 278 Hz and about 275 Hz: two different notes!

Can you hear the difference? Listen!

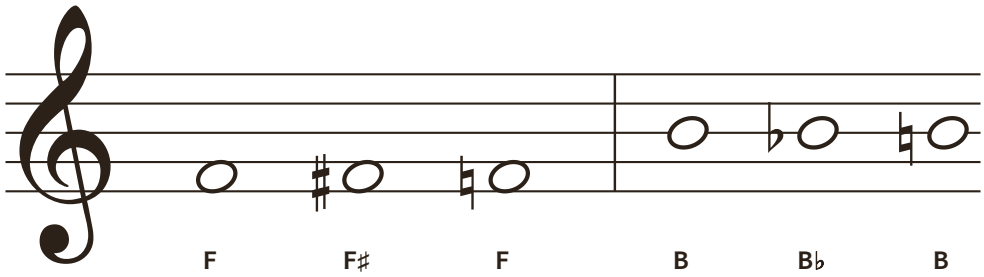
In the Melody Machine, play **The Tower of Fifths**: choose C and stack fifths one on top of the other. With 7 notes you get a scale. Keep going up... after twelve fifths the tower falls over, and then you hear the **comma**!



CANELO SAYS

My name is Canelo, not 'dog': everyone has their own name! It's the same with the steps.

The **sharp** ♯ raises the note by a chromatic semitone (on the piano: an ordinary semitone). The **flat** ♭ lowers it by a chromatic semitone (on the piano: an ordinary semitone). The **natural sign** ♮ rubs out the sharp or the flat: the note is natural again.



Let's play!

1. What does each sign do? ♯ _____ ♭ _____ ♮ _____
2. F♯ with a natural sign becomes _____. B♭ with a natural sign becomes _____.

Did you know...? More than three thousand years ago, in China, people tuned twelve bamboo pipes, each one a fifth higher than the one before. Twelve fifths, twelve pipes: just like the twelve keys in every octave of the piano!



Three different staircases, one music.

CHAPTER 5

What fits between two notes

The chromatic and the diatonic semitone, and the comma



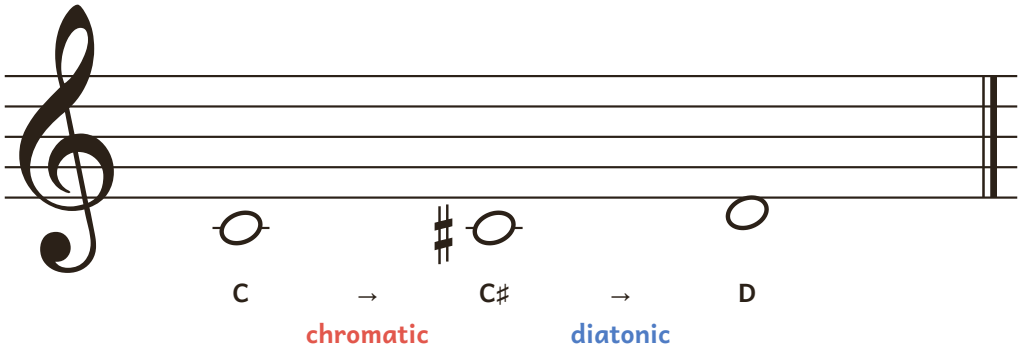
Béla Bartók travels through the villages of Hungary and Romania, recording their songs with a phonograph (from 1906 onwards).

*Listen to how he turned them into music: the **Román népi táncok (Romanian Folk Dances)**, among the Keyboard demos (musicmotionsuite.com).*

A tone, such as C–D, can be split into two pieces. But the two pieces are not the same size, and the difference lies in the **name** of the notes:

Chromatic semitone: the note keeps its name. C → C♯. The note stays in the same place on the stave; it just gets a sharp.

Diatonic semitone: the note gets a different name. C♯ → D, E → F. The note moves to a different line or space.



C → C♯ → D: a tone made of a chromatic and a diatonic semitone.

On the singing staircase, the chromatic semitone is a tiny bit bigger than the diatonic one. That very small difference is called the **comma**. On the piano it doesn't exist: there are only semitones there, all the same size.

That is why C♯ and D♭ are the same key on the piano: they are **enharmonic** notes, two names for the same sound. For the voice and the violin, on the singing staircase, they are two different notes, with a comma between them.

Let's play!

1. Write Ch if it is chromatic and Di if it is diatonic:
   D → D♯ _____ E → F _____ B → C _____ G → G♭ _____ F♯ → G _____
2. Write two ways to get from D to E in semitones:
   D → _____ → E and D → _____ → E
3. Write the other name of the same piano key:
   C♯ = D♭ · D♯ = _____♭ · F♯ = _____♭ · A♭ = _____♯
4. For the voice and the violin, are C♯ and D♭ the same note? _____

Did you know…? The comma is so small that many people can't hear it. But singers and violinists take care of it without thinking: they tune by ear. If you go up twelve fifths from C, you arrive almost at the same C, seven octaves higher… but there is a little piece left over: the comma. The Greeks had already measured it with their monochord.

Let's listen · Hungary, 1906

Béla Bartók and his friend Zoltán Kodály travel round the villages with a **phonograph**, a machine that records sound on wax cylinders. They record country fiddlers and singers who play by ear, and Bartók writes down what he hears. In 1915 he composes the **Román népi táncok**. Listen to the third one, 'Pe loc', which means 'in one place'. The tune repeats an E♯, and at the end an F appears. Which one is higher? The E♯, by a comma!

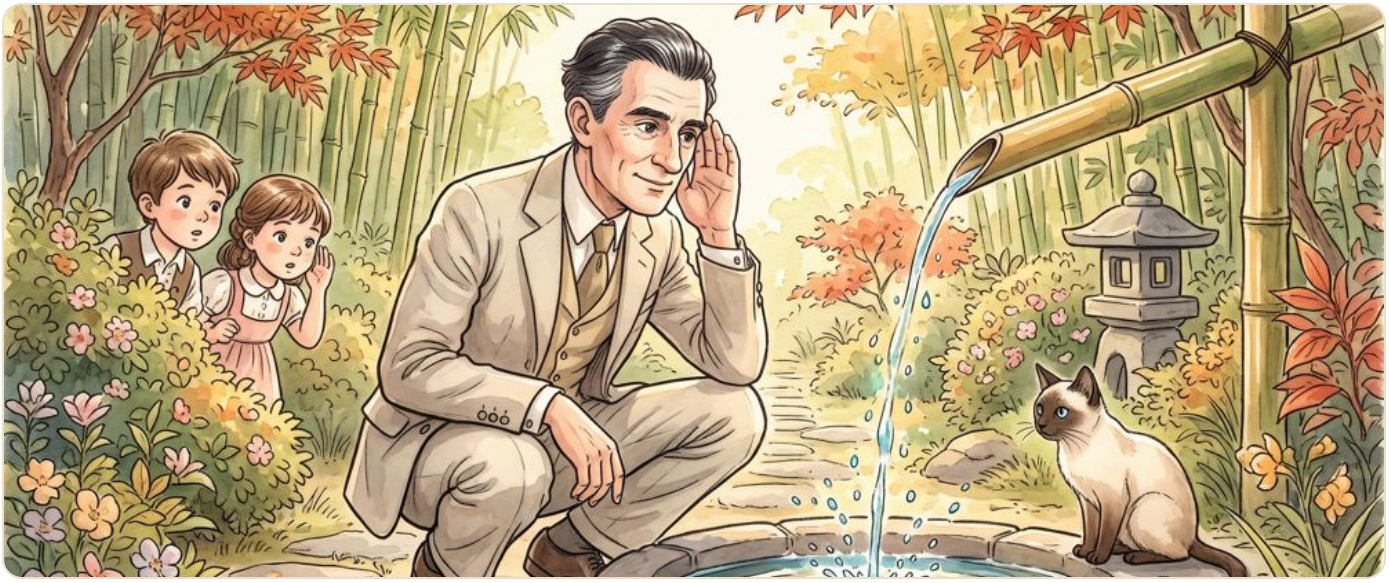
At home

- **Same name or a different name?** One person says two notes ('C, C sharp', 'E, F') and the other answers 'chromatic' or 'diatonic'.
- Ask your grandparents about a song that people used to sing where they grew up. Record it together on the phone, just as Bartók did with his phonograph.

CHAPTER 6

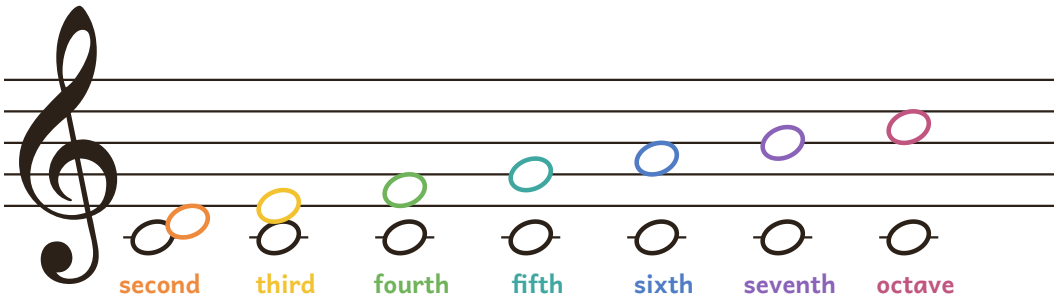
Intervals

The distance between two notes



Maurice Ravel, fascinated by sounds: water, clocks, birds.
*Listen to his **Concerto in G**, 2nd movement, in the Keyboard demos (musicmotionsuite.com).*

An **interval** is the distance between two notes. To name it, you count all the notes, including the bottom one and the top one. From C to E: C, D, E, three notes, *a* **third**. From C to G, five: *a* **fifth**. From C to the next C, eight: *an* **octave**.

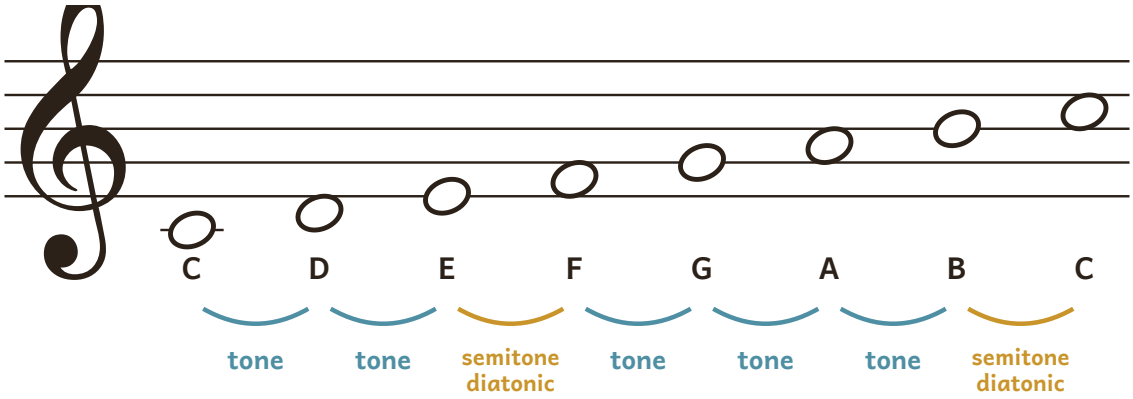


Starting from C: the second (2), the third (3), the fourth (4), the fifth (5), the sixth (6), the seventh (7) and the octave (8).

Let's play!

1. Which interval is it? Count all the notes.
   C–F _____ D–A _____ E–G _____ F–C _____

Two intervals can have the same number and still be different sizes. Look at the steps of the scale of C:



Between E–F and B–C the step is a diatonic semitone; all the others are tones.

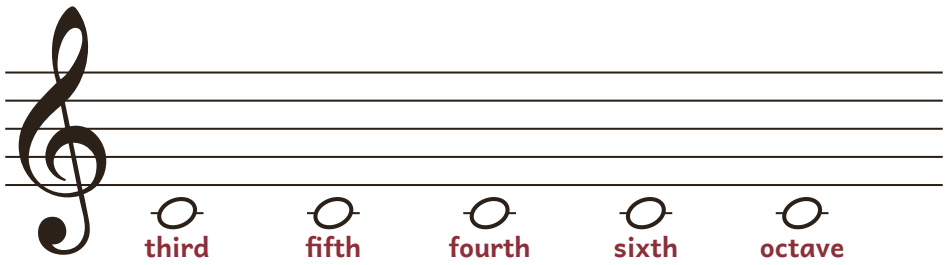
A **major second** is a tone: a chromatic semitone plus a diatonic semitone (C–D). If it has only a diatonic semitone, it is a **minor second** (E–F).

A **major third** has two tones (C–E). A **minor third** has a tone and a diatonic semitone (D–F).

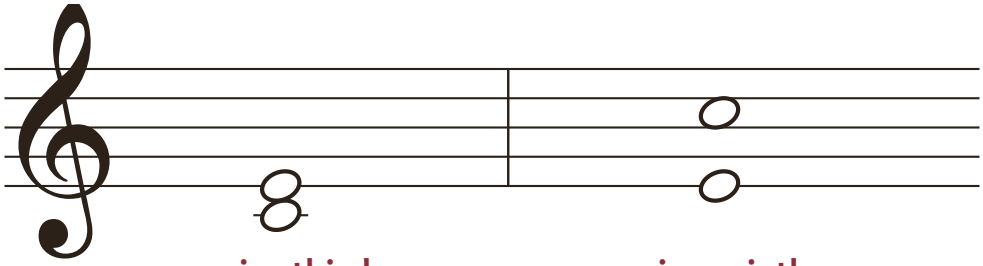
The fourth, the fifth and the octave are called **perfect**: they sound pure, because they are Pythagoras's simplest fractions, 4/3, 3/2 and 2/1.

Let's play!

1. How many of each can you find in the scale of C?
   major seconds _____ minor seconds _____ major thirds _____ minor thirds _____
2. Write a note above each C to make the interval asked for.



To **invert** an interval, you move the bottom note up an octave: C–E becomes E–C. The two numbers always add up to **9**: an inverted third is a sixth (3 + 6 = 9), a second becomes a seventh (2 + 7 = 9). And the size changes: major becomes minor, minor becomes major, and perfect stays perfect.



major third → **minor sixth**

C–E, a major third; inverted, E–C, a minor sixth.

Interval	Inverted
major second (C–D)	minor seventh (D–C)
minor second (E–F)	major seventh (F–E)
major third (C–E)	minor sixth (E–C)
minor third (D–F)	major sixth (F–D)

Let's play!

Turn it upside down: two children come to the front; one is C and the other is E. C climbs over E and now stands at the top. The class counts: E, F, G, A, B, C… a sixth! Try it with C–D and with D–F too.

Interval	Name	Major, minor or perfect	Inverted
C–E	third	major	minor sixth
D–F			
C–G			
C–A			
E–B			

Every interval has its song

Interval	Keys	Starts like…
second	W X	‘Are You Sleeping, Brother John?’: Are you…
third	W C	‘When the Saints Go Marching In’: Oh, when…
fourth	W V	Wagner’s ‘Treulich geführt’ *(Bridal Chorus)*: Here comes…
fifth	W B	‘Twinkle, Twinkle, Little Star’: Twin-kle, twin…
sixth	W N	‘My Bonnie’: My Bon…
octave	W Q	‘Over the Rainbow’: Some-where…

Let’s listen · Paris, 1931

Maurice Ravel finishes his **Concerto in G**; the pianist Marguerite Long plays it for the first time. The second movement begins with the piano alone: a long, calm melody with small intervals, seconds and thirds, like someone singing softly. After 5 seconds it goes from G♯ to A: a minor second! Ravel was looking for the colour of sound, just as the Impressionist painters looked for light.

Did you know…? About three thousand years ago, musicians in India sang their notes as Sa, Re, Ga, Ma, Pa, Dha, Ni. From Sa to Pa is a fifth; from Sa to Ma, a fourth: the same intervals as in Greece, in China and in your school.

At home

On the keyboard, play C (Z) together with each note of the scale: X, C, V, B, N, M and A. Which interval sounds the sweetest? Which sounds the strangest? Sing the beginning of each interval’s song.



CANELO SAYS

In the Melody Machine, play 🐶 **The Echo**: the machine plays and you play it back. That way you hear the intervals and play them yourself. The Cosmic Keyboard has every possible interval: practise them! I can already sing an octave: awooo… AWOOO!

CHAPTER 7

The two clefs

The bass clef: the home of the low notes



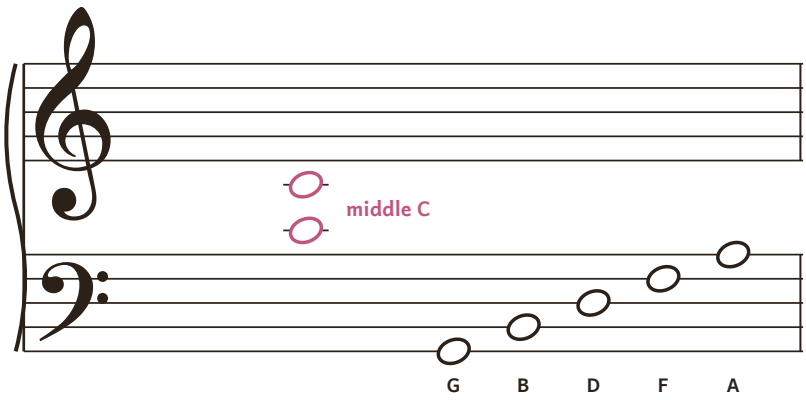
You already know **low** sounds: Dad's or Grandad's voice, the big drum, Mr Mario's double bass. In the treble clef, these notes would fall off the bottom of the stave.



CANELO SAYS

So where do the low notes live? They fall off the stave! They need a home of their own...

Their home is the **bass clef**. When music has a high part and a low part, you write two staves at the same time: that is a **system**. Piano music is always written like this: the right hand on top, in the treble clef, and the left hand underneath, in the bass clef. Between the two lives **middle C**.

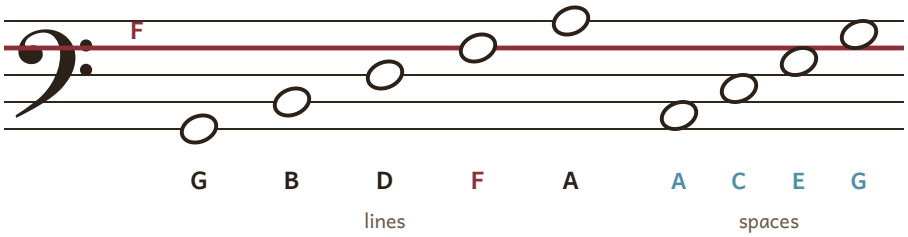


Step by step, with Canelo

1. The bass clef begins with a fat dot and ends with **two little dots**. The dots hug a line: the **fourth**. 'That line is F!' says Canelo. (That is why the bass clef is also called the F clef.)

2. From F I go up and down, step by step, just as in the treble clef: above F comes G, below it comes E.
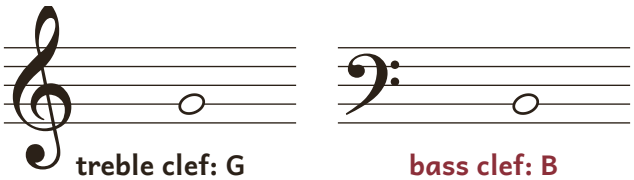
3. Canelo's trick: first look at **which line the clef hangs on**. If it curls round the **2nd line**, that line is **G**. If it hangs on the **4th line**, that line is **F**. 'That's so easy!'





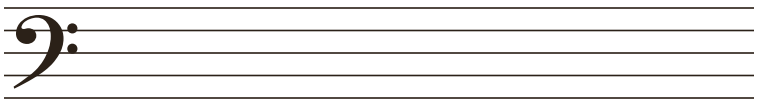
CANELO SAYS

Oops! I read this note as if it were in the treble clef and said 'G'... But in the bass clef it is **B**. First I look at the clef, and only then do I read the note!



Let's play!

1. Write the scale from F to F, with no sharps or flats and without leaving the stave: from the low F (below the 1st line) to the F on the 4th line. You get the **Lydian mode**, a scale with a Greek name. Can you play it on the keyboard? It sounds different because of its B.



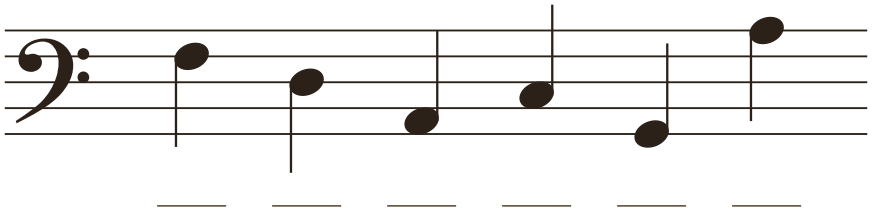
Almost all music has something on top and something underneath: a melody that sings and an **accompaniment** that carries it, with low notes. Listen carefully to what sounds underneath: in the Score, watch the bass-clef stave while the music plays.

Let's listen · what sounds underneath

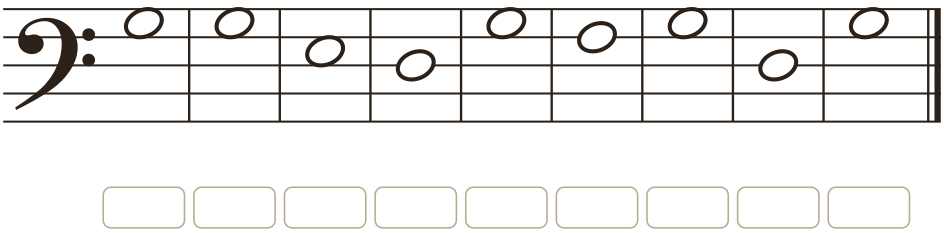
- **The Song of Seikilos** (Greece, 1st century): the **kithara**, a large wooden lyre, plays a long, low note in every bar, the **drone**, while the **aulos**, a double pipe, sings the melody.
- **'Alla Turca'** by Mozart: the piano's left hand keeps the pace, like a drum.
- Schubert's **'Ave Maria'**: the piano carries the voice with a calm movement, like waves.

Let's play!

1. Help Canelo: write the name of each note in the bass clef.



2. This is the kithara's **drone** in the Song of Seikilos: one long note in every bar. Write the name of each note.



3. Trace over the bass clef three times with your pencil, starting at the dot.

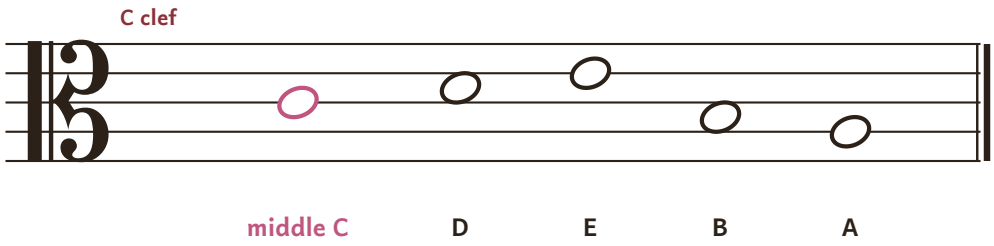
At home

Open the Song of Seikilos in the Score and, while it plays, play the drone on the keyboard, one long key per bar: key G for G · F for F · D for E · S for D. You are the kithara!



Did you know…? About a thousand years ago, monks wrote tiny signs called **neumes** above the words, to remember whether the melody went up or down. Later they put them on lines: first four, then five. To know which line was which, they wrote a letter at the beginning, the name of the note on that line: **G**, **F** or **C**. Over the centuries, those letters turned into drawings: the clefs!

There is also a third clef, the **C clef**: its middle shows where middle C is. The viola uses it, on the third line. On the fourth line it is used by the cello, the double bass, the bassoon and the trombone when they play high.



Let's play!

1. Copy the monks' G and F and, next to them, draw the treble clef and the bass clef.

2. In the bass clef, write the names of the spaces from bottom to top: _____ _____ _____ _____



CANELO SAYS

G, F, C: the clefs were letters! Now I can read the top and the bottom. Woof!

CHAPTER 8

The key to the key

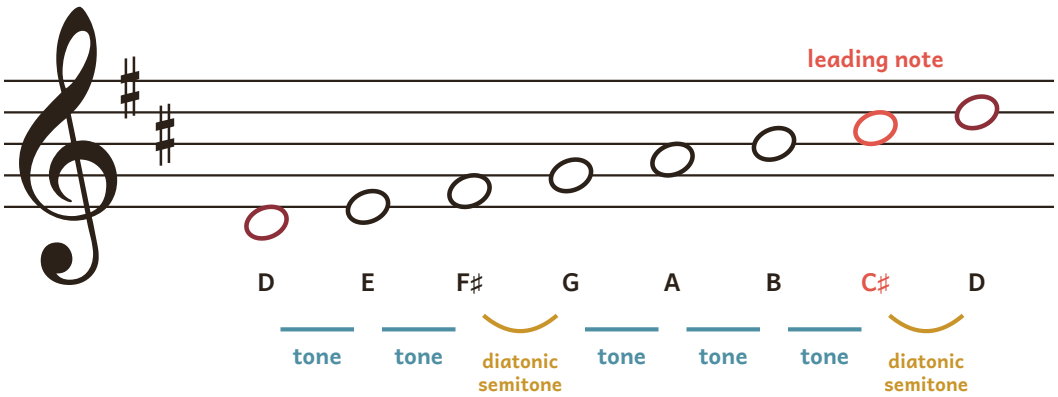
The key signature, the two tricks and the leading note



Every house has its name plate: house D, with two sharps, is D major.

A major scale always has the same shape: tone, tone, diatonic semitone, tone, tone, tone, diatonic semitone (on the piano: a semitone). Starting from C it works by itself, on the white keys. Starting from D you need two sharps: **F♯** and **C♯**.

Instead of writing them every time, you write them once at the beginning, after the clef: that is the **key signature**. It applies to all the notes with that name, right to the end. It tells you which note is home: the **tonic**.
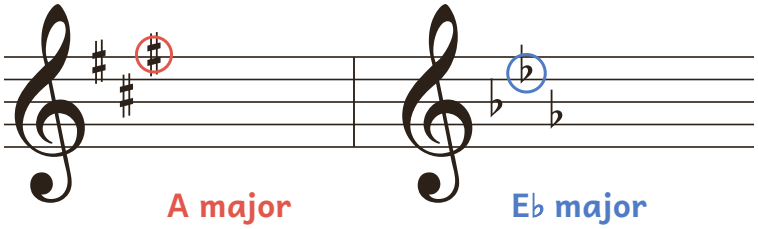


The scale of D major with its key signature: F♯ and C♯. Its formula is the same as for every major scale: ***T T S T T T S*** *(T = tone, S = semitone).*

The sharps and flats of the key signature always appear in the same order and in the same place on the stave. The sharps: **F, C, G, D, A, E, B**, each one a fifth higher than the one before. The flats the other way round: **B, E, A, D, G, C, F**, each one a fourth higher.

The sharps trick. The last sharp is the **leading note**, the note that wants to climb home. The tonic is a diatonic semitone higher. Three sharps: the last one is G♯, so the key is **A major**.

The flats trick. The last flat but one is the name of the key. Three flats: B♭, E♭, A♭; the last but one is E♭, so the key is **E♭ major**. With only one flat there is no last but one: that is **F major**, learn it by heart!



Three sharps: the last one is G♯, and a diatonic semitone higher is A. Three flats: the last but one is E♭.

Let's play!

1. Learn by heart: F, C, G, D, A, E, B. Write it down without looking:

   _____

2. Use the tricks: two sharps → _____ major · one flat → _____ major · two flats → _____ major
3. Now the hard ones: four sharps → _____ major · four flats → _____ major
4. Challenge: on the keyboard, the KEY buttons are in this order: F, C, G, D, A, E, B. Which famous order is that? _____ And if you read them backwards? _____
5. In the **Harmony Machine**, press SPIN and then KEY G, D, A, E: the same melody moves house, and the sharps of the key signature appear one by one, in order.



CANELO SAYS

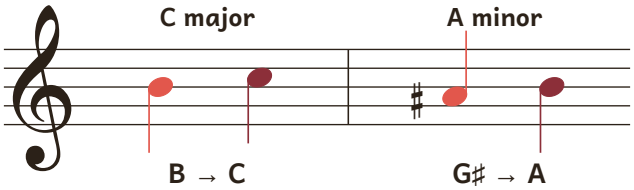
The key signature is like the name plate on the door: before I even go in, I already know whose house it is.

CHAPTER 8 · THE RELATIVE MINOR AND THE LEADING NOTE

Every major key has a minor sister with **the same key signature**: its **relative minor**. It starts on the sixth degree: in C major, on the note A. C major and A minor have no sharps or flats; G major and E minor have an F♯.

Trick: from the major tonic, go down a **minor third**: from C, B, A. A minor! (Or count six notes up: C, D, E, F, G, A.)

And how do you know whether the music is in a major or a minor key? Look for the **leading note**, the seventh degree: it lies a diatonic semitone below the tonic and pushes towards it. In C major it is B. In A minor the scale brings G, a tone below A; the music raises it by a chromatic semitone, to **G♯**. That ♯ is not in the key signature: it appears next to the notes, and it is the clue that the key is minor.



The leading note pushes towards home. Mozart's 'Alla Turca' begins by circling round it: B, A, G♯, A.



CANELO SAYS

I was confused: if the piano only has semitones, why does the piano teacher say 'diatonic semitone'?

Ah, now I get it! She is talking about what is **written**: G♯ → A changes its name, so it is diatonic; but her fingers simply play a semitone. On the piano, G♯ and A♭ are the same key. The violinist, though, plays them differently: his G♯ sits closer to A and pushes harder towards home.

It all depends on the **context**... just like me: at home I am Canelo, and in the park I am 'the little dog'.

Let's play!

1. Write the relative minor: C major → A minor · G major → _____ minor · F major → _____ minor · D major → _____ minor
2. In A minor the leading note is _____ and it rises to A by a _____ semitone
3. In the **Harmony Machine**, press SPIN with KEY C and M: C major, with no key signature. Now press KEY A and m: A minor, with no key signature either! But look at the notes: which sharp or flat appears? _____

Let's play! · Key signature detectives

Download the pieces from musicmotionsuite.com/descargas.html and open them in the Score. Open them one at a time, look at the key signature, use the trick and listen to the ending. Major, or its relative minor?

Piece	Key signature	Key
Beethoven, 'Ode an die Freude'	no sharps or flats	_____
Mozart, 'Alla Turca'	no sharps or flats	_____ (clue: you can hear G♯)
Schubert, 'Ave Maria'	2 flats	_____
Bach, Sinfonia No. 9	4 flats	_____ (clue: it ends on F and you can hear E♮)

Let's listen · Germany, 1722 and 1723

In 1722 Bach completes **Das Wohltemperierte Klavier** (The Well-Tempered Clavier): a prelude and a fugue in each of the **24 keys**, the 12 major and the 12 minor. The following year he makes a fair copy of his **15 Sinfonias** for his pupils, each in a different key: C major, C minor, D major, D minor... None of them has more than four sharps or flats.

Did you know...? The first keyboards were tuned so that some keys sounded very beautiful and others very ugly: in those keys there was a harsh fifth that people called **the wolf**, because it howled. Musicians adjusted the tuning little by little, until today's tuning, which lets you play in every key.

At home

On the keyboard, play J (the leading note, B) and let your hand go by itself to K (C, home). Can you feel that it has arrived home?

Download Bach's 15 Sinfonias in a single .zip: musicmotionsuite.com/suite/demos/bach-sinfonias-bwv787-801.zip. Listen together to two in different keys: which do you like best?

CHAPTER 9

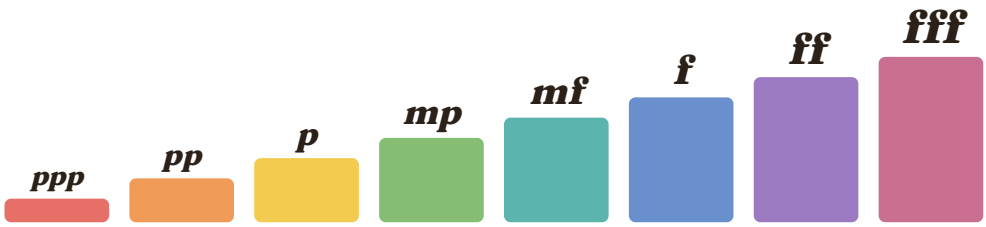
The same notes, different music

Dynamics and articulation



Vienna, 1824: premiere of the Ninth Symphony. Beethoven, by then deaf, conducted; at the end the contralto Caroline Unger turned him round so that he could see the applause.

Dynamic markings tell you how loud the music sounds. They are written under the stave, with Italian letters: **p** (piano) is soft and **f** (forte) is loud. With m (mezzo, 'half') and by repeating the letters, you build a staircase:

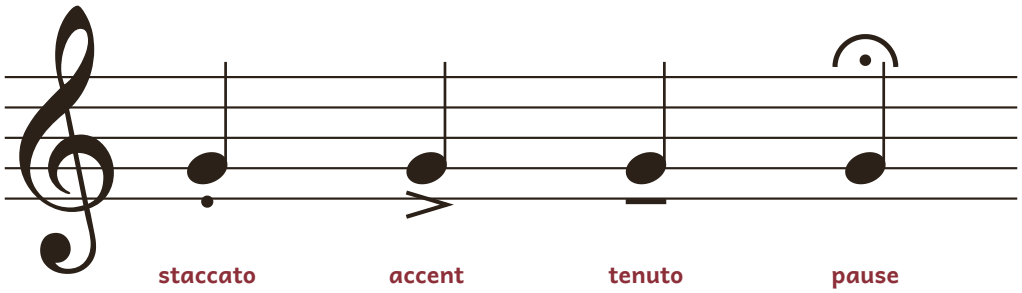


To get louder or softer little by little, there are **hairpins**: the one that opens is the **crescendo**, louder and louder; the one that closes is the **diminuendo**, softer and softer. And the **sforzando** (sf) is a strong accent on a single note; the **fp** (forte-piano) starts loud and drops straight away to piano.

Dynamic markings are not exact numbers: they tell you how one part sounds compared with another. A **p** from the whole orchestra sounds louder than a **p** from one flute on its own. Context again!

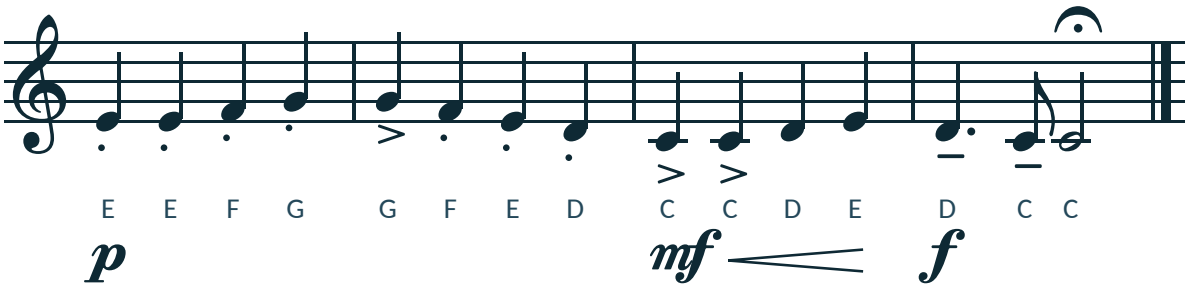
CHAPTER 9 · ARTICULATION

Articulation marks are small signs next to the note. They tell you how the note starts and how long it really lasts.



Staccato (short) · accent (louder) · tenuto (nice and full) · pause, or fermata (the music waits).

This is how they sound in the 'Ode an die Freude': bars 5 to 8, with staccato, accents, tenuto and a pause.



Every instrument articulates in its own way. On the violin, staccato is a very short bow stroke; on the flute, the tongue says 'tu-tu-tu'; for the voice, it is a short syllable; and on the piano, you let go of the key straight away. The sign is the same: every musician translates it for their own instrument!

Let's play!

1. Put them in order, from soft to loud: *f* · *pp* · *mp* · *ff* · *p* · *mf*

2. Draw a crescendo and a diminuendo: _____ _____

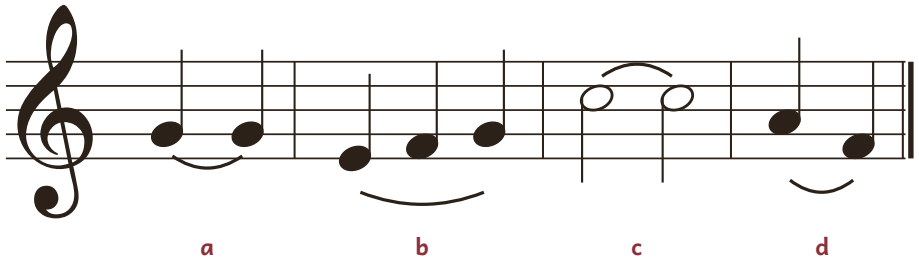
3. Sing G four times, each time with a different articulation: short, loud at the start, nice and full… and the last one until your teacher lowers her hand!



CANELO SAYS

When I give a short bark, woof!, that's staccato. When I howl at the moon, owoooo!, that's tenuto. And when I sit perfectly still waiting for my biscuit… that's a pause!

The **tie** joins two notes that are the same: you play the first one and the sound carries on through both. The **slur** joins different notes: you play them smoothly joined, without breaking the sound (*legato*): on the violin, in one bow; for the voice, in one breath.



The clue: if the two notes have the same name and the curve joins them head to head, it is a tie.

On the violin, the bow goes back and forth: down-bow (⊓) and up-bow (V). Each slur is one journey of the bow; at the end of it, the bow changes direction.

On the flute and for the voice, each slur is one breath. At the end of it, the musician breathes, just like when you drink juice through a straw: you can't suck it all up in one go, you have to stop for breath!

In the 'Ode an die Freude', bars 1 to 4: a slur (legato), a crescendo towards the forte and a diminuendo at the end of the phrase.



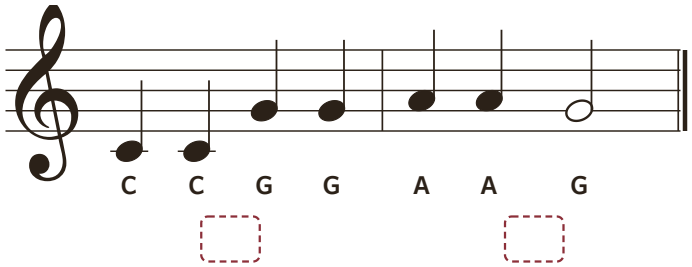
Let's play!

1. Write T if it is a tie and S if it is a slur: a _____ · b _____ · c _____ · d _____
2. With a tie, how many times do you play the note? _____
3. Sing G and hold it for three beats without singing it again: a minim and a crotchet, tied – one G lasting three beats! Then sing E, F, G, A in one breath.

Did you know...? At first, monks only wrote down which way the melody went. Later they invented signs for long and short, and after that for loud and soft: that is how the note values, the ties and slurs, and the dynamic markings you use today were born.

Sight-singing means reading the note names, singing the tune and following the signs, all at the same time: your eyes, your ears and your attention working together.

Let's play! · 'Twinkle, Twinkle' with dynamics



1. Write a dynamic marking, **p** or **f**, in each box, and sing 'Twinkle, Twinkle, Little Star' with the note names, following your own markings.
2. Now put a dot (staccato) or a little dash (tenuto) above some of the notes, and sing it again.
3. Listen to the Ode with dynamics and put your hand up every time the music gets louder or softer.

Listen · Vienna, 7 May 1824

Beethoven gives the first performance of his **Ninth Symphony**. For years he has been almost deaf: he hears the music inside his head. At the end, a choir sings the **'Ode an die Freude'**. Beethoven loved contrasts: suddenly from pp to ff, and sforzandos where nobody expects them.

At musicmotionsuite.com/descargas.html there are two Odes: the usual one and the one with dynamics. Open them in the Score, one after the other: what has changed, if the notes are the same?

Did you know...? At first, the piano was called the **pianoforte**: it was the first keyboard instrument that could play soft and loud, depending on how hard your finger presses. Its name is a dynamic marking! The computer keyboard, on the other hand, doesn't know how hard you press: that is why the Keyboard has the VOL knob.

At home

Play the Ode together on the keyboard (D D F G · G F D S · A A S D · D S S): one of you plays and the other turns the VOL knob for the dynamics. Then play it letting go of each key straight away, and then holding each key down until the next one. Which sounds more cheerful? Record your best version with REC (●) and listen to it with ▶.

FORTE AND PIANO · REVIEW

Loud and soft all over the world

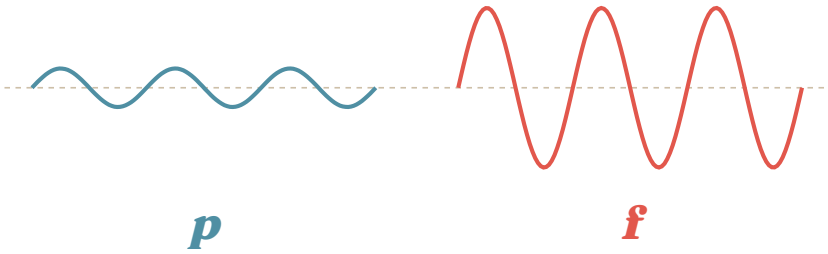
Intensity and volume



On the left, the hand drum of the Indigenous peoples of North America; on the right, the djembe from West Africa. The sound starts pp... and grows to fff!

All over the world, music can be soft and loud. The Indigenous peoples of North America play the hand drum with a soft beater; in West Africa, the djembe is played with the palms of the hands. Both can start almost in silence and grow until the air itself trembles.

When you hit harder, the skin of the drum moves further and the wave in the air gets taller: that is **intensity**. **Volume** is how your ear feels it: more intensity, more volume. But the note doesn't change! With the same number of vibrations per second, only the height of the wave changes.



The same note: soft (p), the wave is low; loud (f), the wave is high.

Let's play!

1. **The drum circle.** The whole class taps the table with their palms, on the same beat. The teacher holds up a card: pp, p, mp, mf, f, ff… and the class changes how loud they play without losing the beat.
2. **The back-to-front echo.** Someone plays a short rhythm in f; the class repeats it in p. Then the other way round.
3. **The VOL knob.** On the keyboard, one of you plays the Ode and the other turns VOL: can you make a crescendo from pp to ff?

Chapter 9 review

1. Write the name of each dynamic marking (pianissimo · piano · mezzoforte · fortissimo): pp → _____ · p → _____ · mf → _____ · ff → _____
2. Which is softer, p or mp? _____ What does a dot above a note do? _____
3. Which curved line joins two notes that are the same? _____
4. With your hands: an **sfz** (one loud clap among soft claps) and an **fp** (loud, then soft straight away).



CANELO SAYS

My ears know everything: with a whispered pp I prick up one ear; with a thundering fff… I hide under the bed! So when you play loud, think of everyone's ears – mine too.

Did you know…? Intensity is measured in **decibels** (dB). A whisper is about 30 dB; a conversation about 60; an orchestra playing fortissimo can go above 100. So always keep your headphones or earbuds at a moderate volume.

At home

Look around the house together for a pp sound (the clock, rain on the window) and an ff sound (the blender, a door slamming). Which would you use to start a lullaby? And for a party?

CHAPTER 10

Chords

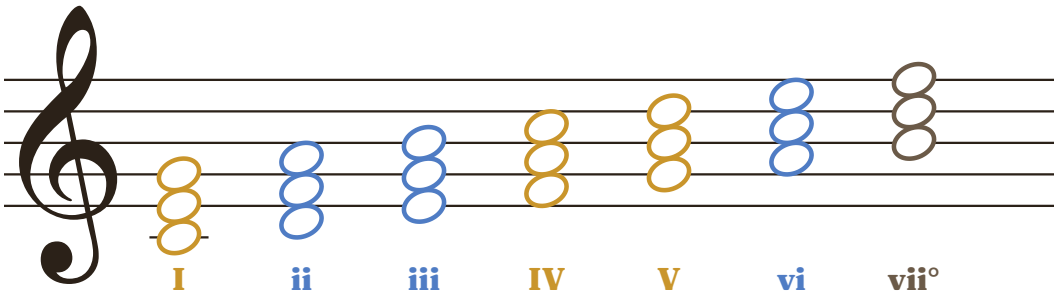
Three notes at once · major and minor · the degrees



*In the Caribbean, the guitar strums chords, and in the songbook they are written with letters: **C · F · G7 · C**. With three chords you can accompany loads of songs!*

A **chord** is three notes sounding together (a triad). You build it from the scale, **take one, skip one**: C (skip D) E (skip F) G. The bottom note gives it its name: it is the chord of C.

If the bottom third is major (C–E), the chord is **major**: bright. If it is minor (D–F), the chord is **minor**: darker, gentler. On each step, or **degree**, of the scale you can build a chord:



*In C major: I, IV and V are major; ii, iii and vi are minor; vii is **diminished**: B–D–F has two minor thirds, and its fifth, B–F, is smaller than the perfect fifth B–F♯: it is a chromatic semitone short.*

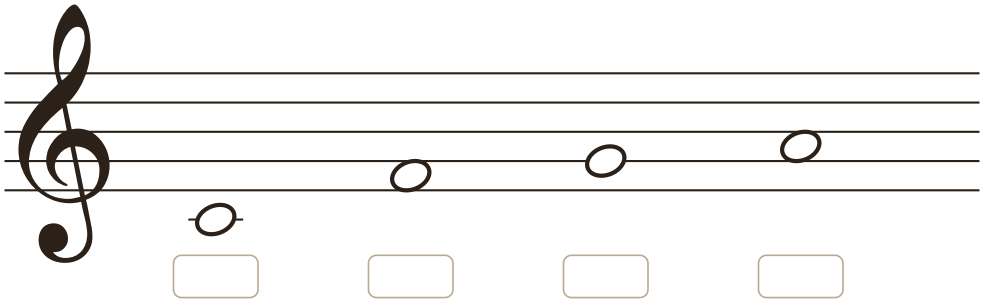


CANELO SAYS

On the stave, a chord looks like a snowman: all three notes on lines, or all three in spaces.

Let's play!

1. Finish each chord: take one, skip one. Write its name and whether it is major or minor.



2. In the picture on the previous page, colour the major chords yellow and the minor chords blue.
3. **Day or night?** The teacher plays a major or a minor chord; the class says 'day' or 'night', with their hand up or down.
4. **Like in the Caribbean:** accompany 'Twinkle, Twinkle, Little Star' with two chords, C major (Z C B) and G major (B M S). C comes at the beginning and at the end; where does G go? Clue: where the tune 'hangs in the air'.

A chord for three. Can't press three keys at once? Three friends, one key each: on the signal, all three together! The same thing happens in a choir: each voice, one note.

Listen · the Song of Seikilos with chords

On each note of the kithara's drone we build a chord: G, G, E minor, D minor, G, F, G, D minor, G. The Greeks did not use chords like these; here we dress up the oldest tune in the world the way a musician of today would.

At home

On the keyboard, play Z C B together: C major. Then X V N: D minor. How does the character change? Ask your child which chord sounds happiest. With the Tonal button, Z C B is always chord I: set MODE to m and the same keys sound minor; set KEY to G and now it is G major.

CHAPTER 11

The chord turns around

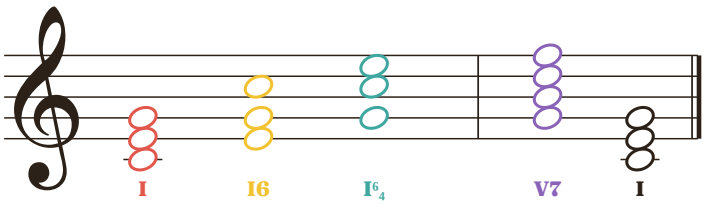
Inversions, chord numbers and the pianist's left hand



*Two ways to accompany. The Scottish bagpipes have a **drone**: their long pipes always play the same note, like the golden ribbon. At the piano, the boy plays the tune and the girl shows the **chords** that change, like little towers of three notes.*

The three notes of a chord can be arranged in three ways, and it is still the same chord. With the root at the bottom, it is in **root position**; with the third at the bottom, in **first inversion**; with the fifth at the bottom, in **second inversion**.

The Roman numeral tells you the degree. The small number tells you which note is at the bottom: no number, the root; **6**, the third; **$^{6}_{4}$**, the fifth.



Chord names with letters. In jazz and pop music, chords are written with the letter of their note: **C** is C major, **D** is D major... A small **m** means minor: **Dm** is D minor. A **7** adds a fourth note: **G7** is G with a seventh, the V7 of C. And the little circle **°** means diminished: **B°**.

You can add a fourth note to chord V, again take one, skip one: G-B-D and, on top, **F**. That is the **V7**. It sounds like a question that wants to go back home, to I.

What are inversions for? If the pianist's left hand played every chord in root position, it would jump all over the place. With inversions, the hand hardly moves.

Let's play!

1. Write the chord number (I, I6 or I^6_4): E-G-C _____ · C-E-G _____ · G-C-E _____
2. Which note do you add to the G chord to make the V7? _____

Did you know...? In the harmony machine, the chord numbers appear under the bass clef. Press PIANO, CLASSICAL and LEVEL 2 · CHORDS, then SPIN and PLAY.

At home

On the keyboard, play A D G (C major), A F H (F major with C at the bottom), M F G (B, F and G) and go back to A D G. Look at your hand: it hardly moves!

Let's play! · With letters

Degree in C major	I	ii	iii	IV	V7	vi	vii°
With letters	**C**	**Dm**	**Em**	**F**	**G7**	**Am**	**B°**

1. Write in letters: F major → _____ · A minor → _____ · G with a seventh → _____ · E minor → _____
2. A song shows: **C · F · G7 · C**. Which degrees are they? _____



*The saxophone sings the tune; from the piano the chords rise up: **C · Am · Dm · G7**.*

CHAPTER 12

We compose an opera

'I've Lost My Sandwich!', in five scenes



The day before the premiere: everyone busy with their own job. And why is Canelo sniffing that school bag...?

In an **opera**, the story is told with words and with music. In ours there is a narrator and there are five scenes, one for each group. At break time, Lucy opens her school bag... but there is no sandwich left in her lunchbox. Each scene has a short dialogue and 8 bars of music that tell us what the characters are feeling.

Scene	Title	Mood	Key
1	Break time	cheerful	C major
2	Where's my sandwich?	worried	D minor
3	The suspicious dog	playful	G major
4	No sandwich	sad	A minor
5	It was here!	happy	C major

Every tune in the opera starts on chord I, wanders through other chords and goes back home, to the home note.

Each group chooses a way. All three end in the same way: the tune starts on chord I, wanders through other chords and **goes back home**, to the home note. Eight bars at ♩ = 88 last about 22 seconds; with a repeat, about 45.

1 · The melody machine

It has four reels, one for each bar. **SPIN** draws notes and rhythms at random; **HOLD** keeps the bar you like; with **KEY** and **MODE** it plays in the key of your scene; **SAVE .mmf** saves it so that you can open it in the Score. For 8 bars, spin twice. A longer act? Spin four times: 16 bars! You'll find it at musicmotionsuite.com/suite/tragamonedas.html.

2 · The dice

One dice for each bar: the number you throw picks one of the six options in the table on the next page. Just like Mozart's dice game!

3 · Our own tune

Make up the tune over the chords: on the first beat of each bar, a note from the chord. Or change the Ode, which you can already play: in **minor** for the sadness, **staccato** and fast for the dog, **ff** with accents for the happy ending.



CANELO SAYS
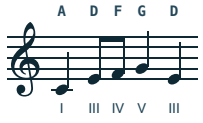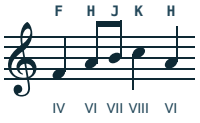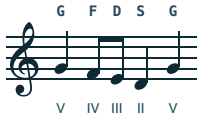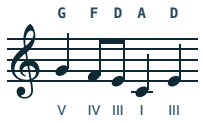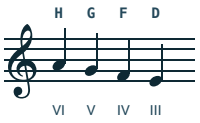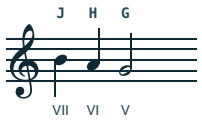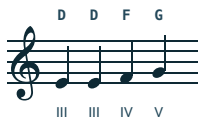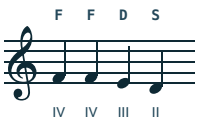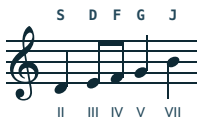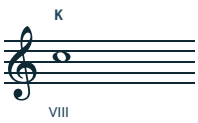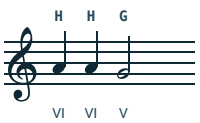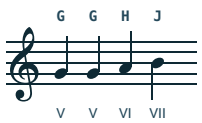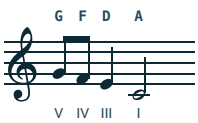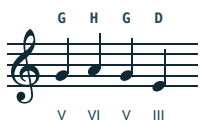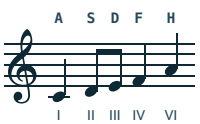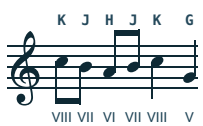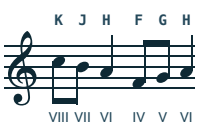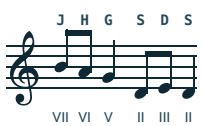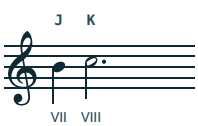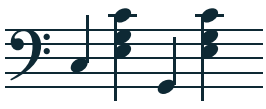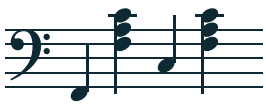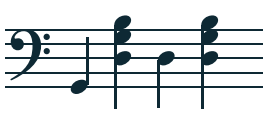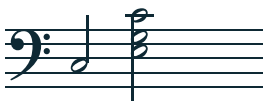
The suspicious dog in scene 3? It wasn't me! Well... ask scene 5.

The time machine · 1792

A publisher brings out a game attributed to Wolfgang Amadeus Mozart, who had died the year before: *composing waltzes with two dice, without knowing anything about music*. For each bar there were eleven possibilities, all on the same harmony, and the dice chose. If the harmony is well thought out, chance cannot spoil it. The melody machine works on the same idea.

CHAPTER 12 · THE OPERA DICE

Roll the dice once for each bar and find the row for that number. Above each note is its key on the middle row of the computer keyboard (with the Tonal button); below it is its degree. At the bottom is the left hand: the same for all six options, because the chord of the bar does not change.

roll	Bar 1 · I	Bar 2 · IV	Bar 3 · V	Bar 4 · I, ending
1				
2				
3				
4				
5				
6				
left-hand				

Let's play!

1. Roll the dice four times and write down: bar 1 → _____ · bar 2 → _____ · bar 3 → _____ · bar 4 → _____. Play it on the keyboard with the KEY and the MODE of your scene.
2. Challenge: three well-known tunes are hidden in the table. Can you find them?

_____

The plan for the 8 bars of our scene. In each bar, write the keyboard letters or the names of the notes; the chord is already there.

Bar	Chord	Our music
1	**I**	
2	**IV**	
3	**V**	
4	**I**	
5	**I**	
6	**IV**	
7	**V**	
8	**I**	

Let's play! · Our scene

1. Our scene is no. _____, in _____. It ends on the note _____, the home note (the tonic).
2. We choose major / minor because _____
3. The dynamics of our scene: _____
4. De dialoog, twee regels:

_____

_____

Did you know...? The dialogue is spoken before the music. But when a character speaks while the music is playing, that has a name in opera: **melodrama**.

At home

Open the melody machine, press SPIN three or four times and listen with PLAY. Use HOLD to keep the bar you like best. Set KEY to A and MODE to m: the same melody sounds sad!

CHAPTER 12 · THE PREMIERE: A SHOW BY EVERYONE

An opera is made by lots of people working together. For our premiere, everyone chooses their job.

Job	What do they do?	Who?
Narrators	They read the story aloud and introduce each scene. (Literature)	
Librettists	They write the dialogues and the words for the programme.	
Set designers	They paint the backdrop for each scene, or project the drawings.	
Costumes	They get the clothes and the props ready: the school bag and the lunchbox with the sandwich!	
Dancers	They dance in the interlude and in the finale.	
Choir	It sings the grand finale.	
Suite technicians	They project the score with the Score (FULL), play live on the keyboard and bring the music we composed to life with the melody machine.	



CANELO SAYS

I've already chosen my job: the suspicious dog. It comes naturally to me!

Let's play!

1. My job at the premiere: _____
2. Draw the backdrop for your scene here, or your character's costume:

Did you know…? In opera, many arts work together, in harmony: **literature** tells the story, **theatre** acts it out, **dance** sets it in motion, and the **visual arts** and the **set** give it its world and its colours. And **music** brings them all together: that is why opera is sometimes called ‘the total work of art’.

I’ve Lost My Sandwich!

An opera by class _____ · date _____

1. **Overture.** The Song of Seikilos with chords: the class ‘kithara’ (chapters 7 and 10).
2. **The opera.** Scenes 1 to 5, each with its own music.
3. **Dance interlude.** Rocinante’s gallop, from Telemann’s suite (looking back at Don Quixote).
4. **Percussion interlude.** The drum circle, from pp to fff (looking back at Forte and piano).
5. **Grand finale.** Everyone sings ‘Ode an die Freude’ *(Ode to Joy)* with dynamics (chapter 9).

Narrators: _____ · Technicians: _____

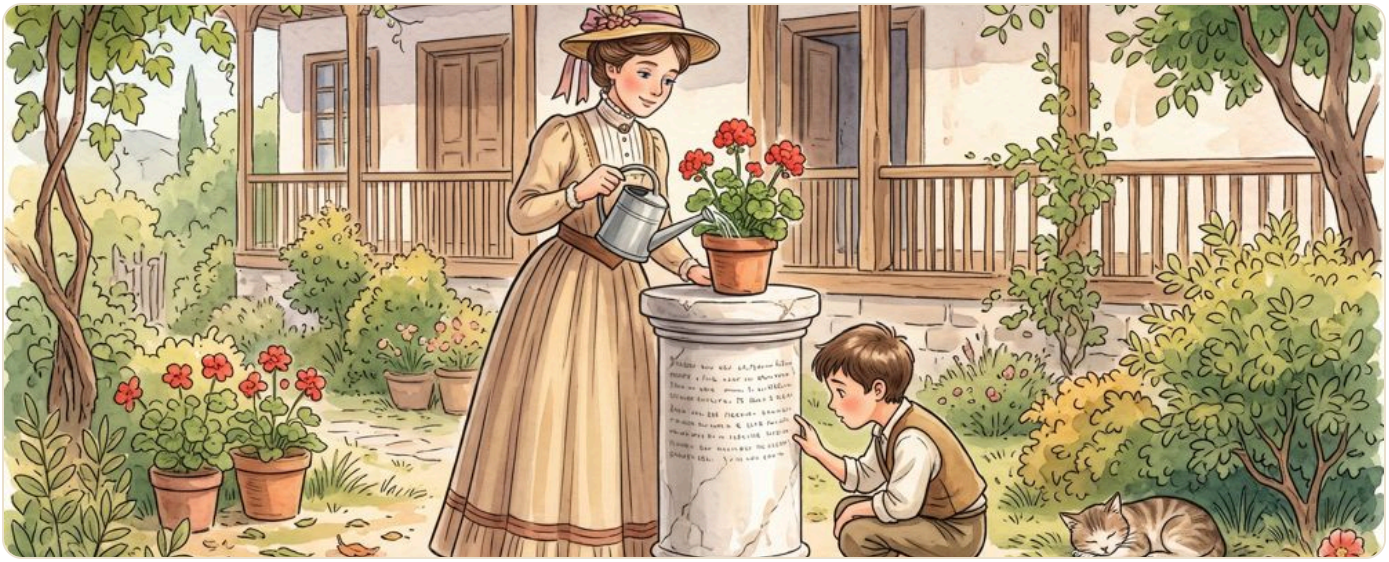


Scene 5, ‘Here it is!’: Lucy finds her sandwich and Canelo breathes a sigh of relief. Applause!

CHAPTER 13

The lost line

The journey of the Seikilos column · how many melodies can there be?



The Seikilos column travelled from Tralles (now in Turkey) to Smyrna, then to The Hague and finally to Copenhagen, where it is today.

You already know the journey of the Seikilos column, from Tralles to the museum in Copenhagen. When its base was sawn off so that a flowerpot could stand on it, the last line of the text was lost… but **the music survived complete**.

How was it written? Above each syllable of the poem there is a **Greek letter** that tells you which note to sing, and some little **dashes** that tell you how long it lasts: just like our note values! That is why the song can still be read, sung and played almost two thousand years later. In the Score you can see each syllable under its note, in 6/8.

Did you know…? The song says: 'While you live, shine; let nothing make you sad: life is short, and time demands its end'.

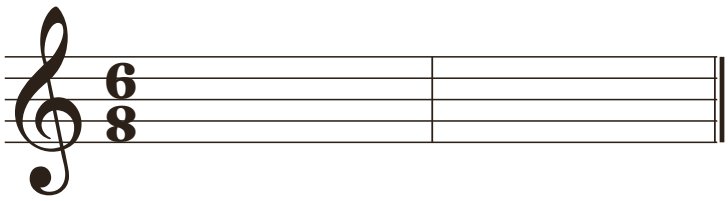
Why we write music down. If the music on the column had been lost, nobody could have guessed it: finding exactly Seikilos's tune among so many possible melodies would be impossible. It survived because someone wrote it down with signs that others could read almost two thousand years later. That is why we keep music, like books, in scores, in archives and in recordings: **what is lost does not come back**.

How many melodies can you write? Let's count only quavers, using the 7 notes from C to B. Each new quaver makes the number of possible melodies **7 times bigger**:

Quavers	Calculation	Possible melodies
1	7	7
2	7 × 7	49
3	7 × 7 × 7	343
6 (one bar of 6/8)	7^6	117,649
12 (two bars)	7^{12}	13,841,287,201
48 (the 8 bars of the Song of Seikilos)	7^{48}	a 37 followed by 39 zeros

Let's play! · My lost line

1. Make up the line that was lost: two bars of 6/8 that end on D.



2. How many melodies do you get with 4 quavers? 7 × 7 × 7 × 7 = _____
3. Challenge: how many melodies of 12 quavers can you make with only the 5 black keys? 5^{12} = _____

Did you know...? About 4.4×10^{17} seconds have passed since the Big Bang. If we had listened to one eight-bar melody every second since then, we would still have heard only a tiny part of all the melodies you can make with quavers alone. There are more possible melodies than stars in the universe!

Dice multiply too:

Game	Calculation	Possible pieces
Mozart: 11 possible bars for each of the 16	11^{16}	about 46 million billion waltzes
The dice machine: 36 squares for each of the 15 bars	36^{15}	as many as there are stars in the universe

Seikilos had his song carved on a column so that it would not be lost. But long before that, music was kept in people's **memory**. In India, hymns were learnt by heart and recited forwards and backwards so as not to make mistakes. In Mesopotamia, clay tablets from almost four thousand years ago explained step by step how to tune a lyre.

Did you know…? Writing is not the beginning of music: it is its memory. Thanks to writing, today you can sing a song from two thousand years ago.

The library of sounds. Every library is full of music, and it only opens for those who can read. In these two levels you have learnt how to open it. Tick what you can already read or do:

- ☐ the stave, the treble clef and the bass clef
- ☐ note values, the dot and the time signature
- ☐ the tone, the two semitones and the comma
- ☐ intervals and keys
- ☐ dynamics and articulations
- ☐ chords and their inversions
- ☐ composing: a bar, a scene, an opera!

Let's play!

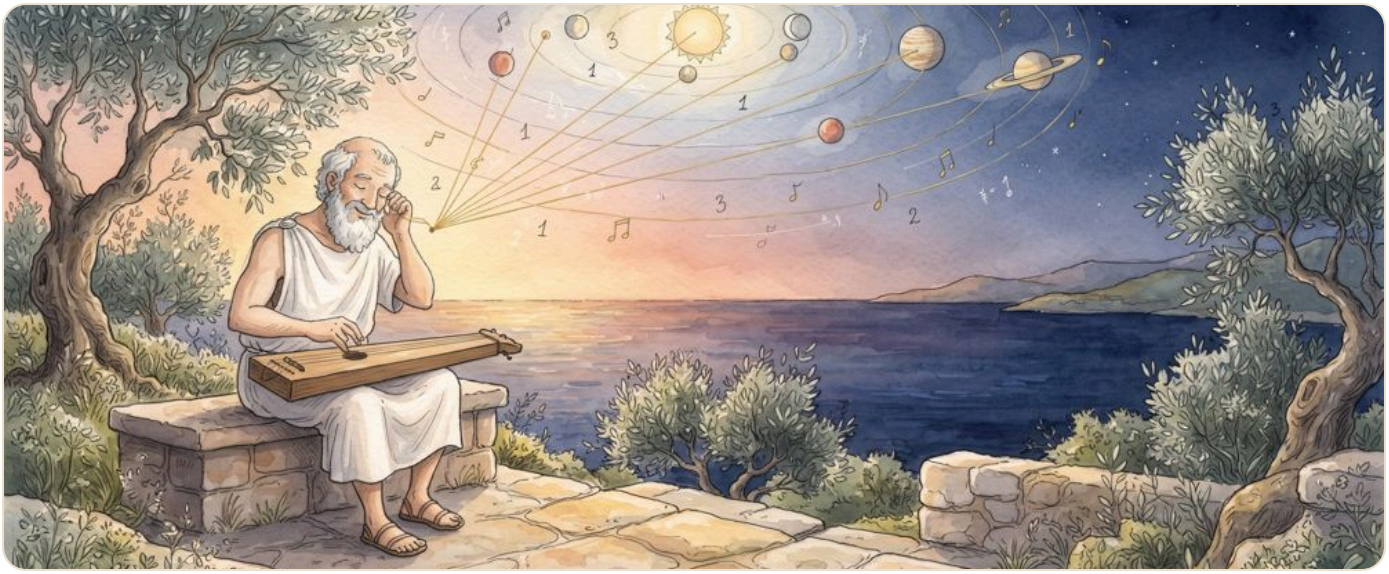
1. How would you keep a song safe so that someone can still sing it in a thousand years' time? _____

At home

Using the calculator on your phone, multiply 7 by itself twelve times together. Ask your child to play their lost line on the keyboard and to tell you what they have learnt to read this year.

Pythagoras and the music of the spheres

Before we roll the dice: where the idea that music is made of numbers comes from.



About 2500 years ago, on the island of Samos, lived **Pythagoras**. One day he noticed something wonderful about a tight string: if he pressed the string exactly in the **middle**, it played the same note, only higher: the **octave**. If he pressed it at **two thirds** of its length, it played the **fifth**, a note that gets on very well with the first one. The sounds that sounded good together were **simple numbers**: 1, 2, 3.

So Pythagoras thought that the whole universe must be made of numbers, and that the Sun, the Moon and the planets made music too as they turned: **the music of the spheres**. Nobody could hear it, he said, because it has been sounding since the day we were born and we are used to it. Two thousand years later, the astronomer **Johannes Kepler** was still looking for that music in the orbits of the planets.

Today we know that the planets don't sing, but Pythagoras was right about something very important: **there are numbers inside music**. In these worksheets we are going to find them:

Worksheets 1 and 2	The dice: why 7 is the king, and how two dice decide how each bar is accompanied.
Worksheet 3	Mozart's game: a die chooses the tune.
Worksheet 4	D, the centre of the notes: a stone in the pond.
Worksheet 5	The numbers of sound: the octave, the fifth and the comma, with a calculator.



Canelo says: I hear music where nobody else does, too... especially when someone opens a packet of biscuits!

Why is 7 the king of the dice?

Two dice, a table and a mountain: this is how the machine decides how to accompany each bar.

1 Two dice, 36 pairs

We roll a **white** die and a **red** one. The white die can show 1, 2, 3, 4, 5 or 6, and for each of those the red die can also show six numbers. That is why there are **6 × 6 = 36** possible pairs. In the table, the white die chooses the **row** and the red die the **column**; each square holds the **sum**.

white dice ↓ \ red dice →	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

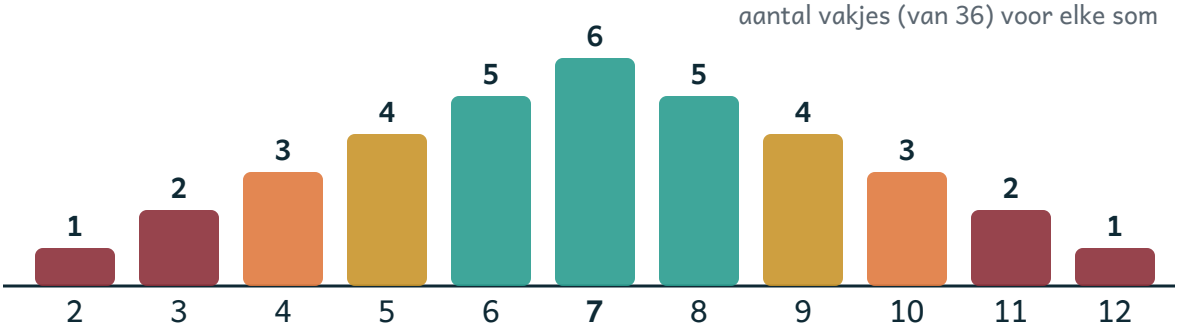
Look at the diagonals. All the squares on the same diagonal ↙ have the same sum. The longest diagonal is the one for **7**: it has **6 squares** (1+6, 2+5, 3+4, 4+3, 5+2, 6+1). The corners, 2 and 12, have only one each.



Canelo says: 7 is the easiest to roll, because it has the most ways to get there!

2 Let's count: the mountain of sums

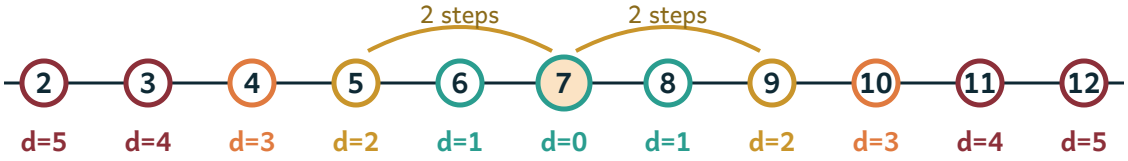
If we count how many squares each sum has and draw it as bars, we get a mountain: it goes up by 1 each time until 7 and then comes back down. 7 is at the top, right in the middle.



In 36 rolls, 7 comes up 6 times (1 in 6); 2 and 12 only once each.

3 How far from 7 did the sum land?

The machine doesn't care whether the sum came out above or below 7, only **how far** away it is. We call that distance **d**. A sum of 5 and a sum of 9 are both **2 steps** from 7.



How the machine plays, bar by bar

The dice don't choose the notes: they choose **how each bar is accompanied**.

4 One roll for each bar

For **each bar**, the machine rolls the two dice **just once**. If your tune has 8 bars, it rolls 8 times. With each roll it does three things:

① It looks at the **white** die and finds its **row** (from top to bottom). ② It looks at the **red** die and finds its **column** (from left to right). ③ Where the row and the column meet, the **colour** of the square says how that bar sounds.

white dice ↓ \ red dice →	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Example: you roll white **3** and red **4**. I go down to row 3, then across to column 4… *a* **green** square! That bar sounds 'as usual'.

Colour	How the bar sounds
green	as usual: **OOM**-pah-pah
gold	the bass jumps: **OOM**-oom-pah
orange	the bass stays put: **OOM**……pah
wine red	the surprise: **OOM**……**OOM**…… (hemiola)

5 A tune of 4 bars: 4 rolls

bar 1	bar 2	bar 3	bar 4
row 3 · column 4	row 1 · column 2	row 2 · column 3	row 6 · column 4
as usual 3 + 4 = 7	**surprise! hemiola** 1 + 2 = 3	**the bass jumps** 2 + 3 = 5	**the bass stays** 6 + 4 = 10

6 Why does green come up so often?

In the table there are **16 green**, 8 gold, 6 orange and only **6 wine-red** squares, in the corners. That is why it nearly always sounds 'as usual', and the surprise comes about **once every 6 bars**. A surprise in every bar would not be a surprise any more!

For the curious · the mathematicians' shortcut. The colours come from the distance to 7: add the two dice, take away 7 and forget the minus sign. You write it like this: **d = | a + b − 7 |**. Distance 0 or 1: green · 2: gold · 3: orange · 4 or 5: wine red. (White 3 + red 4 = 7 → 7 − 7 = 0 → green.)

7 Let's play!

Roll two real dice (one light, one dark) **4 times**. For each roll, find the square and colour the bar in that square's colour:
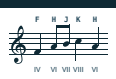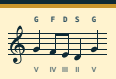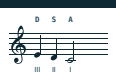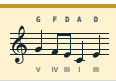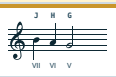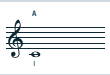
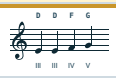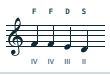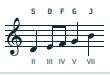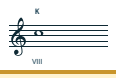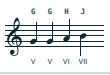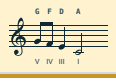
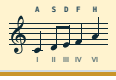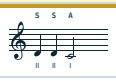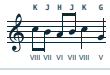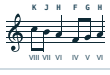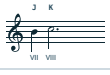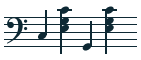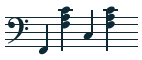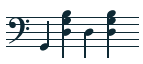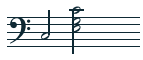
bar 1 · ☐ + ☐	bar 2 · ☐ + ☐	bar 3 · ☐ + ☐	bar 4 · ☐ + ☐

The dice choose the tune

Mozart's game (1792) had **16 bars**, each with **11 options** chosen with two dice. Ours is shorter: **4 bars**, 6 options and **just one die**.

8 A table with 4 columns: one for each bar

The tune has **4 bars**. Each **column** is a bar and has **6 options**, one for each number on the die. You roll **once for each bar**: the number tells you which **row** it is, and you copy out that bar. Four rolls, four bars: your own tune!

die	Bar 1 · I	Bar 2 · IV	Bar 3 · V	Bar 4 · I, ending
1				
2				
3				
4				
5				
6				
left hand				

Above each note is its key (middle row, Tonal button); below it, its degree. Last row: the left hand, the same for all six options.

9 An example: you roll 2, 5, 1 and 4

bar 1 · dice **2** → row 2



bar 2 · dice **5** → row 5



bar 3 · dice **1** → row 1



bar 4 · dice **4** → row 4



10 Why does it always sound good?

Each column is written on **its chord** (I, IV, V and I again), just as in Mozart's game: whichever option you roll, it fits its bar. And the last column always comes back home, to **C**. With 6 options in each of the 4 bars you get **6 × 6 × 6 × 6 = 1296 different tunes**.



Canelo says: three well-known tunes are hidden in the table. Find them!
(Clue: the Ode starts in row 3; 'Twinkle, Twinkle' in row 4.)

The dice of worksheet 2 and the die of this worksheet: the **two** dice choose **how each bar is accompanied** (in the harmony machine); the die of this worksheet chooses **the notes** of each bar. You can use them together: first the tune, then the accompaniment.

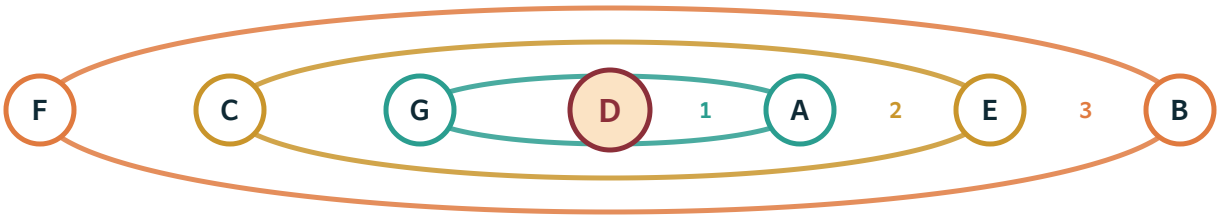
D, the centre of the notes

A stone in the pond, the 7 of the dice and the D on the keyboard: three centres with the same pattern.



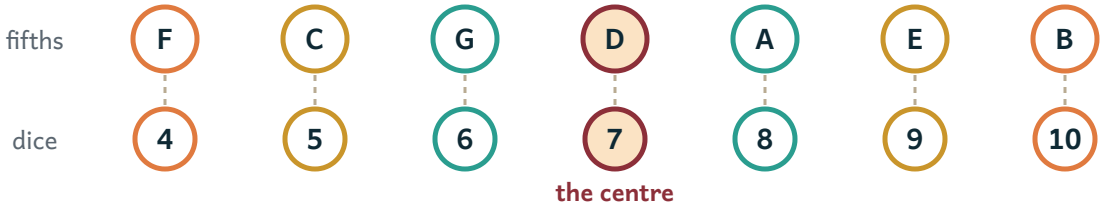
11 The stone in the pond

When the stone falls into the water, the ripples spread out **equally in every direction**: first a small ring, then a bigger one, and another. The natural notes (the white keys) have their stone too: **D**. If we line them up in **fifths**, D is in the middle, with three notes on each side.



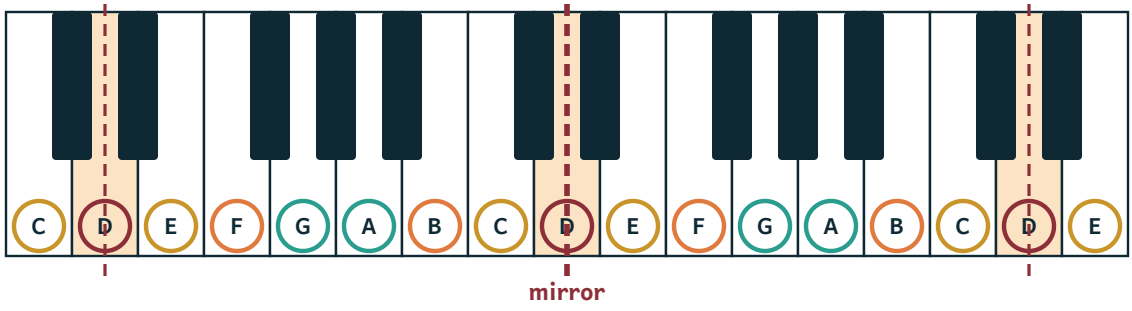
12 Two centres, the same pattern

Put the number line of the dice (worksheet 1) under the line of fifths: **7** and **D** land exactly on top of each other. G and A are **1 step** from D, just as 6 and 8 are from 7; C and E are **2**; F and B are **3**, at the ends. What counts is the distance from the centre, not the side.



13 The keyboard's mirror

Look at the keyboard: **D** sits exactly between the **two black keys**. If you put a mirror on D, the keyboard looks the same on both sides. And if you play the white keys from D to D, the steps are **1 · ½ · 1 · 1 · 1 · ½ · 1** (tone and semitone): they read the same forwards and backwards.



The numbers of sound

Pythagoras discovered it with a string: sounds that sound good together are simple numbers. For this worksheet you need a calculator.

14 The octave is double

A string that vibrates **twice as many times per second** sounds like the same note, an octave higher. The musicians' A vibrates 440 times per second (440 Hz): the octave below is 220, the octave above 880. **Octave = × 2.**

15 The fifth is three halves

Going up a **fifth** means multiplying by **3/2** (that is, × 1.5). If the number goes past the octave, we divide it by 2 to come back home. This is how you build the whole line of fifths, starting from D:

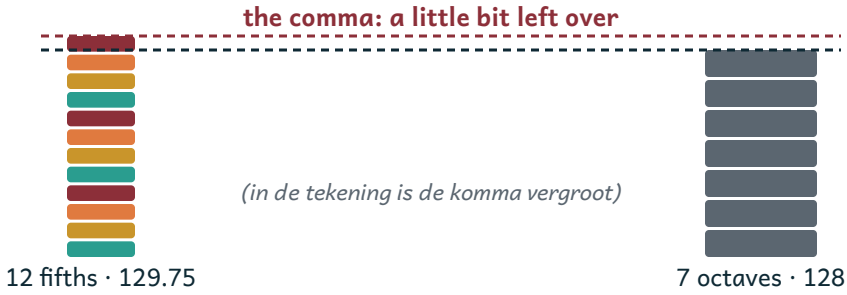
F	C	G	D	A	E	B
347.7	260.7	391.1	**293.3**	**440**	330	495

In Hz. To the right, × 3/2 (A = 293.3 × 1.5 = 440); to the left, ÷ 3/2. Dividing by 2 or multiplying by 2 keeps each note in the same octave.

Work it out: E = 440 × 1.5 = 660 → ÷ 2 = **330**. B = 330 × 1.5 = _____ . G = 293.3 ÷ 1.5 = 195.6 → × 2 = _____ .

16 The comma: the tower that falls over

The melody machine has the game of the **tower of fifths**: each fifth is one more block. With 12 fifths we almost get back to the same note… but not quite. With the calculator:



12 fifths: 1.5 × 1.5 × … (12 times) = **129.75**. 7 octaves: 2 × 2 × … (7 times) = **128**. A tiny bit is left over: 129.75 ÷ 128 = **1.0136**. That tiny bit is the **Pythagorean comma**, and that is why the machine's tower falls over at the 13th fifth.



Canelo says: the 7 on the dice, the D in the notes, the stone in the pond… I can only see one centre: my food bowl!

Answers: B = 495 · G = 391.1 (195.6 × 2).

Music words

A little dictionary to revise Level 2

Upbeat (anacrusis)
The notes that come before the first full bar, like a little hop before you start.

Repeat marks
Two little dots beside the bar line: whatever is between them, you play twice.

Coda
Italian for ‘tail’: the special ending of the piece.

Semiquaver
Half a quaver: four of them fit into one crotchet beat.

Demisemiquaver
Half a semiquaver: eight in one beat!

Dot
A dot next to the note value: the note then lasts half as long again.

Triplet
Three notes in the time of two: ‘tri-pe-ta’.

Ornaments
Little notes that decorate the tune: the trill, the mordent, the appoggiatura.

Hertz (Hz)
How many times a sound vibrates in one second. The tuner’s A: 440 Hz.

Semitone (piano)
The smallest step on the piano: from one key to the key right next to it, white or black.

Chromatic semitone
The note goes up or down and keeps its name: C → C♯.

Diatonic semitone
The note gets a different name: C♯ → D.

Tone
A chromatic and a diatonic semitone together. On the piano: two semitones.

Comma
The tiny difference that turns up after twelve fifths. The piano shares it out and hides it.

Enharmonic notes
Two names for the same piano key: C♯ and D♭. With the voice and on the violin they sound a tiny bit different.

Interval
The distance between two notes. You count both of them: C–E is a third.

Major and minor
Two sizes of the same interval: the major third (C–E) is bigger than the minor third (D–F).

Perfect
The fourth, the fifth and the octave: they sound so pure that they are neither major nor minor.

Inverting an interval
The bottom note moves to the top. The two numbers add up to 9: a third becomes a sixth.

Bass clef (F clef)
Gives line 4 the name F. It is the home of the low notes.

C clef
The line the clef sits on is C. The viola uses it.

Drone
A long, low note that keeps sounding the same, as on the bagpipes or the kithara.

Key signature
The sharps or flats at the start of every stave: they count for the whole piece.

Key
The home of the music: the home note and the scale, major or minor.

Tonic
The home note: that is where the tune begins and ends.

Leading note
The seventh degree, a diatonic semitone below the tonic: it pushes towards home.

Relative minor
The little sister of a major key: the same key signature, and it starts on the sixth degree. C major → A minor.

Dynamics
How loud the music sounds: pp, p, mp, mf, f, ff.

Crescendo and diminuendo
Crescendo: louder and louder. Diminuendo: softer and softer.

Sforzando (sf)
A strong push on one single note. With fp you start loud and drop straight away to piano.

Articulation
How a note begins and how long it lasts: staccato (short), accent, tenuto (full length), pause or fermata (the music waits).

Tie
Joins two notes that are the same: you play the note once and it sounds for both.

Slur
Joins different notes without breaking the sound: that is legato.

Sight-singing (solfège)
Reading the names, singing and following the signs, all at the same time.

Intensity and loudness
Intensity is the strength with which the sound vibrates; loudness is how the ear feels that strength. They are measured in decibels (dB).

Chord
Three notes at once, skipping one each time: C–E–G.

Degree
Each step of the key, in Roman numerals: I, ii, iii, IV, V, vi, vii°.

Diminished chord
Two minor thirds stacked on top of each other, like B–D–F: its fifth is smaller.

Inverting a chord
The same three notes, with a different one at the bottom: I, I6, I^6_4.

Chord figures
The number that shows the degree and says which note is at the bottom. V7 adds a fourth note.

Chord symbols
The letters used in pop music and jazz: C, Dm, G7, Am.

Opera
A story told with music, acting, dance and scenery.

Libretto
The words of the opera: what the characters say.

Melodrama
A character speaks while the music plays.

Overture
The music that opens the show, before the story begins.



CANELO SAYS

Forgotten a word? Look it up here... and then go back to the chapter to see it in action. I revise them too!



I composed my opera!

My Little Music Grammar · Level 2

This diploma is awarded to

_____

who travelled from Seikilos to Ravel, read in the treble clef and the bass clef,
built chords, got to know the keys
and composed a scene of the opera with the group.

_____ _____

Teacher Date